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Gaze Interaction in Modern Trucks

av

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Linköpings universitet

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Södertälje June 2014

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Abstract

In this master thesis project carried out on Scania's interaction design department in Södertälje an evaluation of the technology gaze interaction has been done. The aim was to see if the technology was suitable for implementation in a truck environment and what potential it had. The work started by doing a context analysis to get a deeper knowledge of the research done on within the area related to the subject. Following the context analysis a comprehensive need finding process was done. In this process, data from interviews, observations, ride along with truck drivers, benchmarking and more was analysed. The analysis of this was used to identify the user needs. Based on the user needs the concept development phase was conducted. The whole development phase was done in different stages and started off by an idea generation process. The work flow was made in small iterations with the idea to continuously improve the concepts. All concepts were evaluated in a concept scoring chart to see which of the concepts that best fulfilled the concept specifications. The concepts that best could highlight the techniques strengths and weaknesses were chosen and these are Head Up Display Interaction and Gaze Support System.. These concepts focused on the interaction part of the technique rather than a specific function. Test of the two concepts were conducted in a simulator to get data and see how they performed compared to today's Scania trucks. The result overall was good and the test subjects were impressed with the systems. However there was no significance in most of the cases of driving except for some conditions where the concepts prove to be better than the systems used today. Gaze interaction is a technology that is suitable for a truck driving environment given that a few slight improvements are made. Implementation of the concepts have a good potential of reducing road accidents caused by human errors.

Keywords: Eyetracking, eyetracker, gaze, gaze interaction, simulator, simulator tests, Scania, truck, trucks, LTU, LiU, Luleå University of Technology, Linköping University, user centred design, user focused design, user friendly,

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1 Introduction

1.1 Project Incentives

The transport industry is an increasing and essential part of our infrastructure. Without heavy goods transports our society would come to a complete stop. During 2011, Swedish registered trucks carried out 35,2 million haulage assignments and a total of 325 million tonnes were transported only in Sweden (Trafikanalys, 2012). A typical workday for a long haulage truck driver consists of loading and unloading cargo and driving long distances, all within a strict time limit. This brings a lot of stress into the work environment. At the same time many drivers often experience being tired or under stimulated during their working day. These factors provide danger that does not belong in a traffic situation. Of all the road accidents that occur each year, about 20% involves collision with a heavy truck (Hjort & Sandin, 2012). The technological advancements have made it possible to counter these problems in new and interesting ways that could result in monetary gain and saved lives.

Modern trucks are equipped with an increasing amount of smart systems that enable the driver to interact in new ways with their vehicle, other drivers and their surroundings. With the increased amount of functionality also comes higher demands on driver safety and a well designed user interface. The driver needs to manoeuvre all the different functions of the truck while still keeping focus on the road and this must be possible in a safe and reliable way.

User interaction in a truck has up until now mainly consisted of the driver interpreting information after which the driver reacts through physical contact using levers and buttons. New technology however opens up for new and innovative ways for the driver to interact with the system. Gaze interaction enables the drivers to interact with their vehicle using only their eyes. The eyetracking technology can also be used to observe the driver and to act as a safety system if the driver for some reason is unable to drive the vehicle in a safe way. Ideally gaze has the ability to let the driver focus on the core activity; driving.

Today Scania is one of the world's leading manufacturers of heavy trucks and buses. Industrial and Marine Engines is another important business area. Scania has been a part of the Swedish industry since 1891 and is dedicated to building the best possible trucks and to have a high quality product delivered to its customers. With tough competition it is important to always stay at the forefront of technology and driver safety. Scania initiated this project in order to investigate if gaze interaction is a feasible technology for the truck of the future and if it will keep Scania in its current leading position as a company of quality and safety.

1.2 Stakeholders

This project contains a number of different stakeholders that are described below. The report is primarily directed towards Scania and the examining partners at the respective universities. It is also relevant to anyone who is interested in the gaze interaction technology, the service design methodology or driver interaction and interaction design.

1.2.1 Core Stakeholders

The core stakeholders in this project are the authors and their supervisors that have a great influence on the direction of the project and its decision making.

1.2.2 Primary stakeholders

The target user group is the focus of this project and also one of the primary stakeholders. More specifically it is the drivers, who are directly affected if this system is implemented, that are a part of the primary stakeholder group. The drivers that have taken part in the development of this project through interviews, workshops and tests also have a direct impact on the project outcome and quality. The two universities; Linköping University and Luleå University of Technology are also part of this project as they are responsible for the quality of their educations and are responsible for the examination of the thesis authors. The Driver Vehicle Interaction department at Scania started this project and has great interest in the outcome to see if the technology is suitable for future generation trucks.

1.2.3 Secondary stakeholders

The secondary stakeholders that are involved in this project are the supplier of the gaze interaction technology, Tobii. The project depends on the technology that they deliver even though they are not strictly part of the development work. Society in general is also a stakeholder in this project. If the proposed solutions are implemented in the transport industry then society will be affected through safer transports.

1.3 Aims and objectives

The aim of this project is to investigate the gaze interaction technology and to assess if it is a technology that is suitable for the truck driver environment. This will be done by studying the target user group before developing one or more innovative concepts that are then tested and evaluated in the Scania driving simulator.

1.3.1 Research questions

In order to fully understand the area of interest the following questions will need answering:

1. *What information is possible to extract from the driver using eye tracking while driving on the highway?*
2. *How does a gaze interaction interface affect the driving experience?*
3. *Can gaze interaction utilise the peripheral field of view in order to optimise the information flow?*
4. *Is gaze Interaction a technology that is suitable for the truck driver environment and interface?*

1.4 Project scope and limitations

The project spans over 18 weeks from the beginning of February until the middle of June. The amount of work corresponds to 30 credits sums up to a total of 800h per person. The main focus of the study is long haulage highway driving.

The time and budget of this project did not allow for concept testing on real roads. Instead the entire driver testing took place in the Scania driving simulator which has the advantage of high repeatability. No considerations were taken to manufacturing costs, further development costs or spare equipment around the test rig. The only thing that was budgeted was the purchase of the actual gaze equipment and the consultant from Tobii who helped with the installation and training.

There was only one person working full time with the simulator and help with testing and setup was limited to his working hours. In a way the concepts were limited in how advanced they could be due to what was possible to test in the simulator and the time given in the simulator. The time had to be shared between other research projects and simulator renovations, all with their own agenda and priorities.

The gaze interaction technique itself has some limitations regarding the functionality in different light conditions. It works poorer if it is really dark or really bright. Furthermore is the recordable field of vision limited as well, both in height and in width. The gaze interaction concepts in this project only include the interaction between the vehicle and the driver and does not include any other functionality.

Since all of the concepts had to be tested and evaluated in the driving simulator, which holds current technology, they had to be implementable in the near future. Technology that is not yet tested in a truck i.e. the head up display (HUD) had to be augmented onto the projected image of the simulator. Other than that the focus was on concepts in the near future and little time was given to ideas that were considered too far into the future.

2 Theoretical framework

In order to develop solutions that are relevant and well motivated a theoretical framework is needed. This chapter contains literature relevant to the decisions that have been made during the course of the project.

2.1 Research Theory

This chapter contain relevant research in the fields of gaze interaction, human vision and Interface design guidelines.

2.1.1 Human Vision

To create one or more powerful concepts with gaze interaction, one needs to understand how the human vision works and how the product can use the strengths and weaknesses to its advantage. Using this information it is easier to understand where to present important information and what types of interaction that should be preferred over other.

How the Human Vision Works

Each of the human eyes have approximately six million retinal cone cells. The cells are more tightly packed in the centre of our visual field, an area called the fovea. The fovea is only approximately 1% of the retina but occupies about 50% of the visual cortex's input (Tobii Technology, 2010). To understand how small the fovea is compared to the entire human visual field the comparison could be done by extending your arm and look at your thumbnail. Everything that is not your thumbnail falls outside the fovea. The area outside of the fovea is the peripheral field of view. The visual field inside the fovea is extremely high, for a healthy human often higher than in modern cameras. Outside this region the resolution quickly drops down to a few dots per centimetre viewed at an arm's length. At the very edge of our peripheral vision field the pixels are as large as melons compared to a thumbnail inside the fovea (Johnson, 2010).

The impression of seeing the whole world in full focus is only an illusion by our brain. This is accomplished by moving our eyes very quickly and filling in the rest of the information with what we already know and expect. Our brains do not need a high resolution image of the world around us since it is easy to sample and resample information when needed (Tobii Technology, 2010) (Johnson, 2010).

Our eyes moves from one fixation to another and the time when the eyes move in between these fixations is called a saccade. During a saccade our eyes do not register any visual information but the time is often too short to notice.

One might wonder why we have our peripheral vision if the fovea is better in every way. The peripheral vision is extremely good at scanning our environment for visual

clues that the fovea then can focus on. When searching for orange capsicums in a grocery store our peripheral vision helps us draw attention to anything that is orange even if it is capsicum or oranges. The peripheral vision is also extremely sensitive to motion. Anything that moves even slightly in our periphery is likely to draw attention to it. This was especially useful to our ancestors that could find food or detect predators in their environment. The eyes can move both by consciously pointing them somewhere or by reflex, very fast. Since the brain fills in the information that is missing, a spot with information that does not draw attention to it will end up never being seen without us even noticing (Johnson, 2010).

Field of View

Human beings have an ambinoocular field of view that spans 188 degrees from left to right. Ambinoocular means the combined visual field of the left and right eye together. The upper and lower limits to our field of vision is 50 and 70 degrees respectively from the standard sight line. The standard sight line is a horizontal line projected from the centre of our eyes straight forward. In order to see the controls on an instrument panel the sightlines are more limited and only span about 60 degrees from left to right and top to bottom. It is recommended to place any visual controls within this region in order for them to be accessed and perceived comfortably. The field of view for emergency control is even more limited and only occupies 30 degrees of the visual field left to right and top to bottom from the standard sight line (Diffrient, Tilley, & Harman, 1993). Any important object outside these regions might not get detected. The human field of view is illustrated in Figure 1.

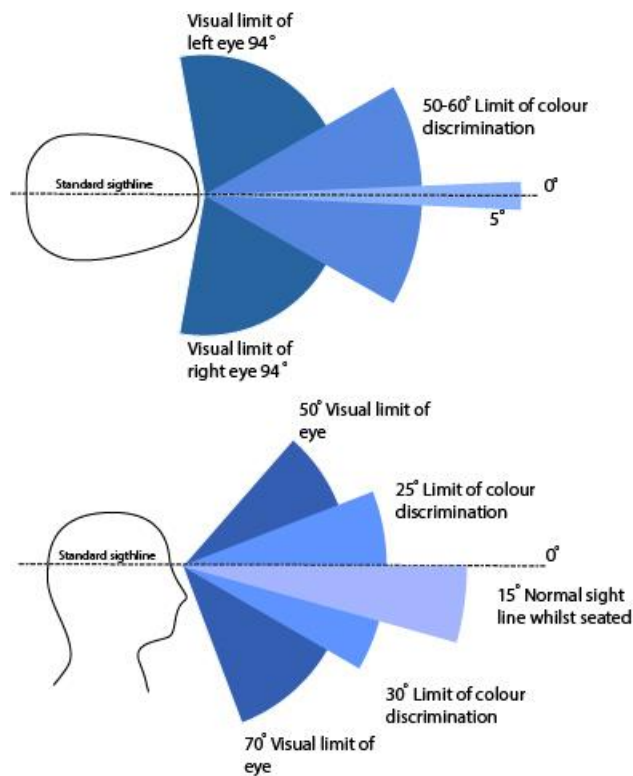


Figure 1 - Human Visual Limits Adapted from Diffrient et. al.

Safe Driving

The American institute of National Highway Traffic Safety Administration (NHTSA) states that glances away from the forward road scene for more than 2,0 seconds greatly increases the crash or near-crash risk. When the drivers glance away from the forward road for more than 2,0 seconds in a 6-second period, the risk of an unsafe event increases substantially relative to normal driving (National Highway Traffic Safety Administration, 2012). This implies that any information or task that draws the driver's attention from the road increases the likelihood of a crash.

The time between two saccades is usually called a fixation duration. This event is closely related to cognitive processing in alert subjects but has failed to show an unequivocal relationship to sleepiness. Fixations shorter than 150ms and longer than 900ms are closely associated with lower cognitive processing. When a subject is tired the ratio of express and overlong fixations increases. They both seem to increase the same amount which indicates that the driver, when sleepy, loses interest in the driving environment (Schleicher, Galley, Briest, & Galley, Blinks and saccades as indicators of fatigue in sleepiness warnings: looking tired?, 2008). This means that a tired driver that fixates for too long on a specific point on the road is more likely to end up in a road accident.

2.1.2 User Interface

Designing an interface is a complex and big process due to the amount of different aspects to take into account. There are many different sources and opinions when looking at rules or guidelines so there are many things to have in mind when designing an interface (Nielsen J. , Usability engineering, 1994) (The Commission of the European Communities, 2007) (Norman, Design principles for human-computer interfaces., 1983). Nielsen states that the system needs to match the real world. "The system should speak the users' language, with words, phrases and concepts familiar to the user, rather than system-oriented terms. Follow real-world conventions, making information appear in a natural and logical order." Another example of an important aspect Nielsen brings up is the visibility of system status, which means that the user should always be informed of what is going on through "appropriate feedback within reasonable time" (Nielsen J. , Usability engineering, 1994).

Driving Interface

When driving a vehicle it is important to always be able to keep at least one hand at the steering wheel at all time, while interacting with the interface. Information with higher safety relevance should be prioritized ahead of everything else. When designing an interface the aim should be to have everything that is going to be interacted with mounted in a good and accessible place. In this way the user can reach everything with ease and see relevant information (The Commission of the European Communities, 2007).

Alert Messages

There are some common methods on how to get a user's attention. Firstly one should place information where the user is looking, preferably within the field of vision. Any message should be marked, and since humans read symbols faster than text it is recommended to have a symbol. A symbol also has the advantage of being read by people that do not speak the same language, something text could not do. To really get a user's attention, information can be displayed as a pop-up message or wiggling sign. Our vision is designed to perceive movements and our focus is quickly drawn to a flashing area. To further enhance the message urgency the information can be complimented by using sound to alert the user that something is happening. Drawing attention in this manner should be reserved only for very important messages. The human brain pays less and less attention to stimuli that occur frequently, this is called *habituation* (Johnson, 2010).

UX

User experience (UX) is described by Roto et. al. as a subset of experience as a general concept, since it is related to the experiences of using a system. UX is unique to each individual and is influenced by prior experiences and expectations based on those experiences. UX is not the same as usability even though usability typically is an aspect contributing to the overall UX. It is important to know that the UX is dependent on the context. Even though the system is the same, the context may change, altering the UX of that system. A system can also contain properties

connected to the brand and its values or if changes are done to the product e.g. change the background picture of a phone or scratches that makes a device look old and worn. UX Design (UXD) is similar to Human Centred Design (HCD) when it comes to the importance of involving users in the iterative work process. UX however adds important dimensions to the challenge of implementing HCD in a mature form (Roto, Law, Vermeeren, & Hoonhout, 2011).

User Feedback

Shneiderman speaks about the need for user feedback among other things. He states that it is important to always give relevant feedback of what is going on to the user. It is also important to strive for consistency in an interface in order for the users to recognise and learn the system more quickly. To balance the feedback given to the user is of importance, for example frequent and minor actions can have a modest response while infrequent and major actions should have more substantial response. The use of shortcuts is often a desired feature for frequent users, also Abbreviations, function keys and hidden commands are helpful for expert users (Shneiderman, 2004).

2.1.3 Gaze Interaction

Gaze interaction is a technology where one uses their eyes to interact with an interface. The idea is that the user does not have to use their hands or any other unnecessary movement while interacting (Kern, Mahr, Castronovo, Schmidt, & Müller, Making Use of Drivers' Glances onto the Screen for Explicit Gaze-Based Interaction, 2010). This is especially useful for disabled people that are limited to how much they can interact with their body and drivers that cannot let go of the steering wheel (Morimoto, Flickner, & Amir, Free Head Motion Eye Gaze Tracking Without Calibration, 2002). The difference between eye-tracking and gaze interaction is that eye-tracking only registers the position of the eyes where gaze interaction allows for a two-way communication where the user also can interact. The system can calculate where a person is looking and if a person who is driving a car is unfocused or tired. Gaze interaction can also be used for interaction i.e. by controlling functions or other things in an interface.

Technology

The Tobii eye trackers work through a technology that is called Pupil Centre Corneal Reflection (PCCR). The idea is that the user's eyes are lit up by an infrared light creating reflections that are then captured and measured. A vector is then calculated with the angle between the reflection and the cornea. This vector combined with the geometrical features from the reflections enables the device to calculate the user's gaze (Tobii Technology, 2010). The eye tracking device works at a distance between 40-90 centimetres at an accuracy level of up to 0,4 degrees in ideal conditions (Tobii Technology, 2014). A schematic on how the eye tracking technology works can be seen in Figure 2.

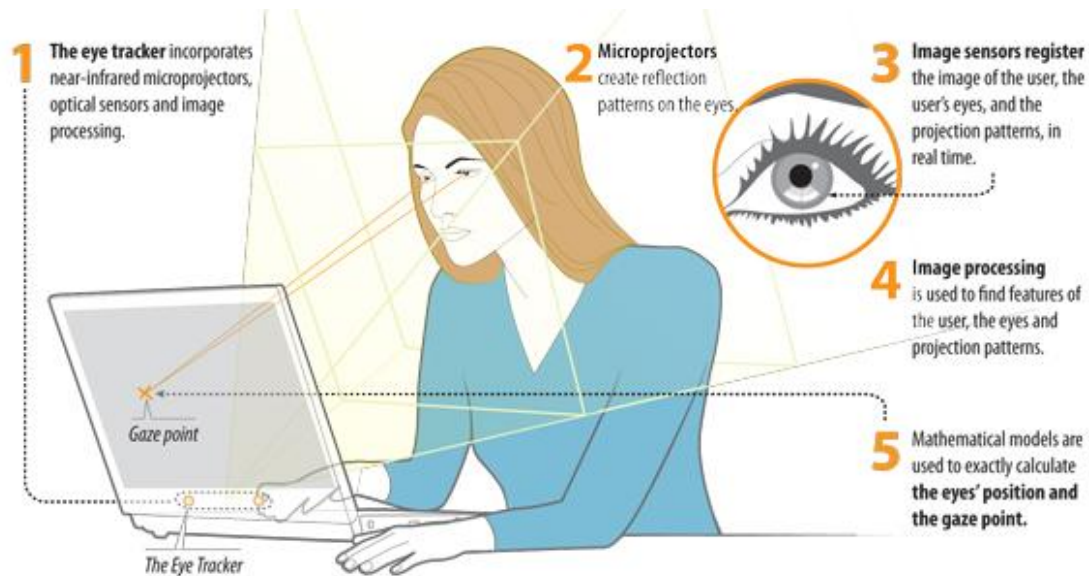


Figure 2 - The Eye tracker functionality, Tobii Technology 2013

The data that can be extracted from the eye tracking device is the following:

- Eyetracker timestamp
- Validity of left and right eye individually ranging 0-4
- 3D coordinates of left and right eye (X,Y,Z)
- 2D coordinates of left and right eye (X,Y)
- Pupil diameter of left and right eye.

The data is recorded and stored in a data file for later analysis. The validity code indicates the system's confidence in correctly identifying which is left and right eye for each specific sample. There are a number of factors that can affect the accuracy of the eye tracker. Among them are eye movements, the calibration procedure, drift and ambient light.

Drift

Drift is a gradual decrease in the eye tracking accuracy over time, compared to the true eye position, and can be affected by a number of factors. One cause could be changes in the eye physiology such as degree of wetness, tears or variations in the environment i.e. sunlight variations. Drift problems however only occur if the eye tracking sessions are very long or if the test conditions rapidly and radically change. Tobii claim that their eye tracking products cope well with drift even though extreme changes in the test environment will produce a significant drift effect (Tobii Technology, 2010).

Ambient Light

The ambient light has a significant impact on the measured accuracy and precision (Tobii Technology, 2014). The measurement accuracy is the closeness of agreement

between a measured value and the true value of what is measured. The precision of a measurement is the closeness of agreement between indications or measured values obtained by replicate measurements on similar objects under identical conditions (Joint Committee for Guides in Metrology, 2008). The accuracy of the eye tracking device can be viewed in Figure 3 and the precision in varying illumination is demonstrated in Figure 4.

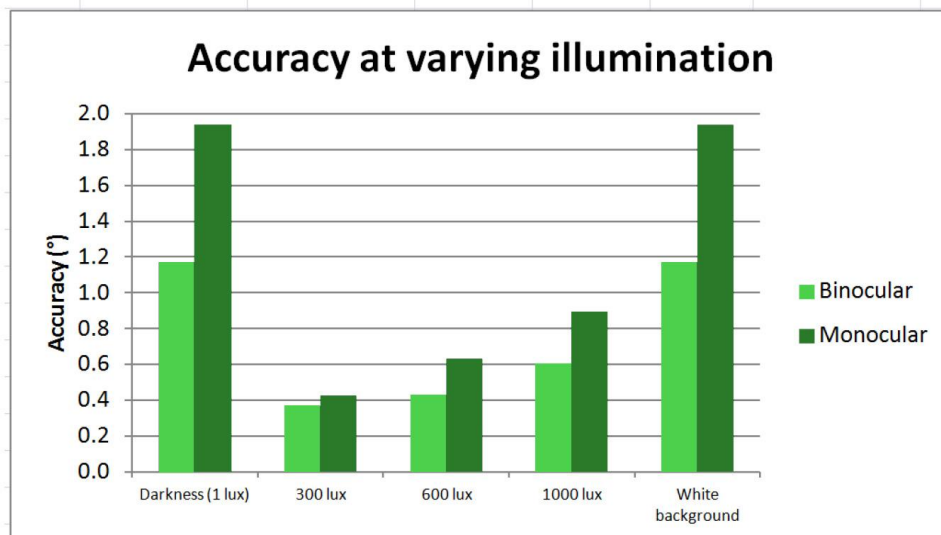


Figure 3 - Accuracy at Varying Illumination, Tobii Technology 2013

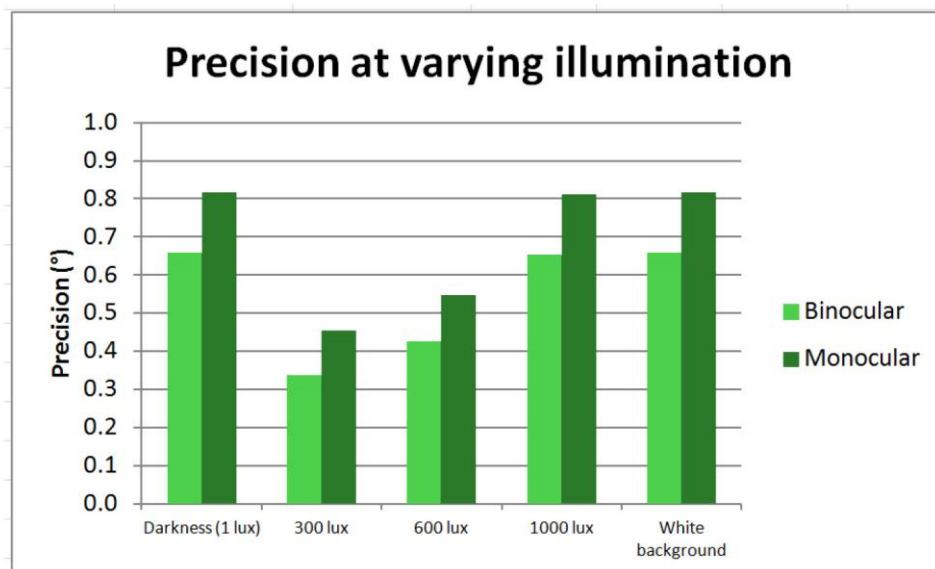


Figure 4 - Precision at Varying Illumination, Tobii Technology 2013

Benchmarking

There is a lot of research going on at an academic level on gaze interaction. Volvo cars have presented a concept that is not yet ready to be put into production (Volvo Cars, 2014). Fraunhofer IDMT have developed a smaller and cheaper eye tracking solution that does not require any calibration. That system however is not as powerful as the Tobii eye tracking solution (Fraunhofer Institute for Digital Media Technology, 2014). When this report is written there is no driver system in trucks or cars that use gaze interaction to monitor and improve the driver experience on the market. However, there are many indications that extensive research is being conducted at various companies and that potential products could be launched within a few years.

Toyota has what is called a driver attention monitoring system that identifies when the driver's face is not directed towards the road. If the driver is not looking at the road then the car lightly presses the brakes and if a crash is imminent, then the seat belt is retracted and the brakes are primed to reduce the impact force (Toyota, 2008). This system was introduced in 2006 and many car companies have released their own systems since that time. A literature search indicates that a lot of driver attention research has been made.

Previous Work

Most of the research done today consists of a large variety of different approaches on how to use gaze. Some researchers have looked on how to use it just when it is needed, if for example a user focuses on one thing on an interface that thing will enlarge in order for the user to see it better (Miniotas, Špakov, & MacKenzie, Eye Gaze Interaction with Expanding Targets, 2004). Others have made research if it is a good compliment to a mouse or touch when using a computer (Sibert & Jacob, Evaluation of Eye Gaze Interaction, 2000) (Kern, Mahr, Castronovo, Schmidt, & Müller, Making Use of Drivers' Glances onto the Screen for Explicit Gaze-Based Interaction, 2010). Other things that have been covered in research is whether the use of a single camera (to track the eyes) can be implemented with a good outcome. Research shows that the calibration times in these studies are reduced significantly (Hennessey, Nouredin, & Lawrence, A Single Camera Eye-Gaze Tracking System with Free Head Motion, 2006). Much of the research done is to find new ways of using the gaze interaction and one thing that is important when trying to find new areas to use gaze is to make it easier to use and implement. (Ohno & Mukawa, A Free-head, Simple Calibration, Gaze Tracking System That Enables Gaze-Based Interaction, 2004) (Morimoto, Flickner, & Amir, Free Head Motion Eye Gaze Tracking Without Calibration, 2002) (Hennessey, Nouredin, & Lawrence, A Single Camera Eye-Gaze Tracking System with Free Head Motion, 2006).

Midas Touch

Controlling computers through gaze interaction can provide a fast and efficient method of interacting with computers (Huckauf, Goettel, & Heinbockel, What you don't look at is what you get: Anti-saccades can reduce the Midas Touch-problem, 2005) (Penkar, Lutterith, & Weber, 2012). The challenge however lies not in the technology but in the type of interaction that is suitable for this (Jacob, 1990). The term Midas touch was originally coined, in the gaze sense of the term, by Robert J.K. Jacob in 1990 and has been frequently used since then. Midas touch is when everything the user looks at gets activated even if the user does not intend to. At first it might be empowering to activate functions just by looking at them. But before long people get annoyed by accidentally activating everything they look at. Normal visual perception requires that the eyes move around and then focuses on an object before action (Jacob, 1990). This means that a user tends to first look around between the different options before looking at the final selection before action. There are a number of suggested solutions to how the Midas touch problem is to be solved. Penkar et al suggests that a method where selection is controlled by letting the eye dwell on a certain spot on the screen. The test subjects performed simple tasks such as answering questions by looking at the correct answer. They also suggest that the results were improved by moving the answers away from the actual button. This reduced the problem with Midas touch significantly. Furthermore it is recommended to have an anchor point on the buttons for better accuracy (Penkar, Lutterith, & Weber, 2012).

Another solution is the one suggested by Huckauf et al. By using anti-saccades and letting the user control their selection not by looking directly onto the selected object but rather looking at an area next to it they managed to reduce the dwell time. However the error rate was much higher and more research is needed (Huckauf, Goettel, & Heinbockel, What you don't look at is what you get: Anti-saccades can reduce the Midas Touch-problem, 2005). A similar but better approach is proposed by Istance et al. They propose what is called the snap clutch approach. The idea is to switch between different selection modes by moving your eye out of the screen in different directions. By doing this the user can easily switch off the dwell gaze interaction that causes the Midas touch problem. The technology works well in a 2D space and is easy to implement on almost any application (Istance, Bates, Hyrskykari, & Vickers, Snap Clutch, a Moded Approach to Solving the Midas Touch Problem, 2008).

Surakka et al propose yet another method where the user looks and frowns as a trigger for the gaze interaction. The interaction technique shows great potential in aiding physically challenged people with the only drawback that it requires electrodes to be attached to the user's face in order for it to work (Surakka & Illi, Gazing and Frowning as a New Human-Computer Interaction Technique, 2004). Lastly Bednarik et al use an approach where they use computer learning to distinguish when a user has the intent to click or when they are just looking. Since the field of research is still young, it is not possible to process the data in real time.

The technology however may very well be interesting in the future as a complement to existing data input methods (Bednarik, Vrzakova, & Hradis, What do you want to do next: A novel approach for intent prediction in gaze-based interaction, 2012).

Hand Eye Coordination

As shown there are many solutions to dealing with the Midas touch issue but there is one type of interaction that has gained more wind than the others. That is to combine gaze and hand interaction. Stellmach et al describe this as “gaze suggests and touch confirms” (Stellmach & Dachzelt, Look & touch: gaze-supported target acquisition, 2012). Turner et al has a similar approach to this type of interaction. A typical action would be to hold down two fingers on a touch display and thereby activating a drag and drop functionality. The object would follow the users gaze until the fingers were released and the object would stop where the user stopped with their gaze. By doing this Turner et al propose an intuitive interface that allows users to manipulate object out of reach for normal touch as well as working around the Midas touch issue (Turner, Bulling, & Gellersen, Combinig Gaze with Manual Interaction to Extend Physical Reach, 2011). In the same manner Stellmach et al used a touch device to confirm what the eye was looking at and received very positive results. This shows that hands and eyes working together is a powerful way of operating a gaze supported system (Stellmach & Dachzelt, Look & touch: gaze-supported target acquisition, 2012).

2.1.4 Head Up Display (HUD)

A Head Up Display (HUD) is an instrument which allows the driver of a vehicle to view key information simplified on the windshield in the drivers visual field rather than looking down at the instrument panel for necessary or useful information which have been the traditional way of displaying information. HUD was first used in fighter aircrafts and they were engineered to help the pilot focus attention forward and adapt to the ambient light level in the primary visual field. This will in the end help ease the workload (Weihrauch, Meloeny, & Goesch, 1989).

HUD's consists of a transparent display projected from a small projector that shows information to the driver while still looking forward on the road. Most often a HUD only displays limited and critical information such as speed, turn signal and fuel symbols. General Motors (GM) was the first company to introduce a HUD which was solely developed for automobiles in 1988. Since that it have just evolved and HUD's can be find in many different vehicles from a variety of manufactures. The HUD is generally projected at bumper depth or beyond (Garrett, Bret, & Zeljko, Evaluating the Usability of a Head-Up Display for Selection from Choice Lists in Cars, 2011).

2.1.5 Strong Concepts

A strong concept is described by Höök and Löwgren as more specific than theories on interaction design but more generic than a specific interaction design solution. A strong concept can consist of partial ideas and elements of design. For an intermediate-level design solution to be considered a strong concept it needs to fulfil a few requirements:

- Interactive behaviour rather than static appearance
- It needs to be an interface between technology and people
- It needs to have a core design idea that cuts across particular design situations
- Needs to be more abstract than a specific design solution

Höök and Löwgren talk about social navigation as an example of a strong concept. Social navigation refers to users making decisions based on the decisions of other users. This can be identified for example in the design of websites where “people who bought this also bought...” or on websites where the user has to move through a large set of information i.e. scholar.google.com and www.imdb.com. The information about other people’s choices affect the user and the user’s choices in turn affect other users (Höök & Löwgren, Strong Concepts: Intermediate-Level Knowledge in Interaction Design Research, 2012).

2.2 Product Development

2.2.1 Generic Product Development

Ulrich and Eppinger state that when developing new products it is good to have a standardised process for a number of reasons. The process quality can be assured as well as resources can be coordinated. The planning makes sure that everyone knows what and when to deliver their work. By comparing the development process to the actual events in the project, project managers can easily verify that the project is on track. Furthermore by documenting each project, improvements can be transferred to coming projects making sure that mistakes are not repeated (Ulrich & Eppinger, 2008).

A generic product development process follows six individual steps from the planning phase onto the production ramp-up.

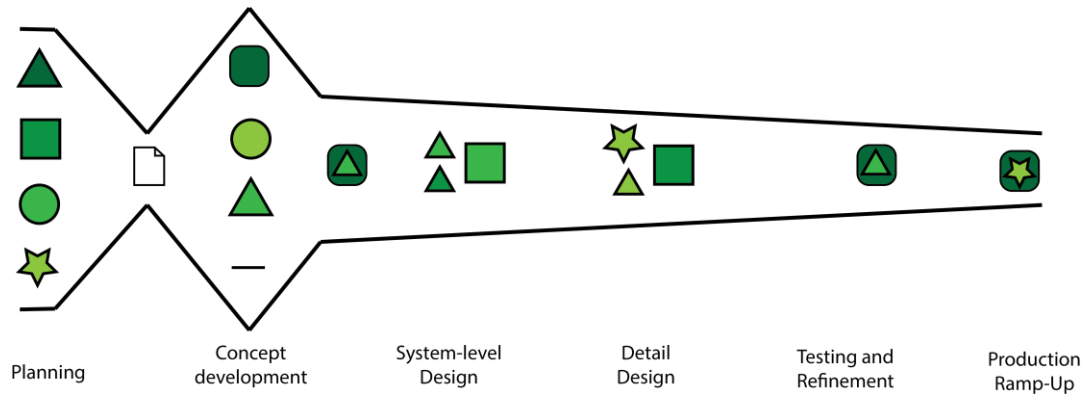


Figure 5 - Generic Product Development Process Adapted from Ulrich & Eppinger

The two initial phases (planning phase and the concept development phase) that take place immediately after the project has been initiated, can be compared to the double diamond often used in service design. The double diamond consists of two expanding and converging phases (Ulrich & Eppinger, 2008) (Stickdorn & Schneider, 2013).

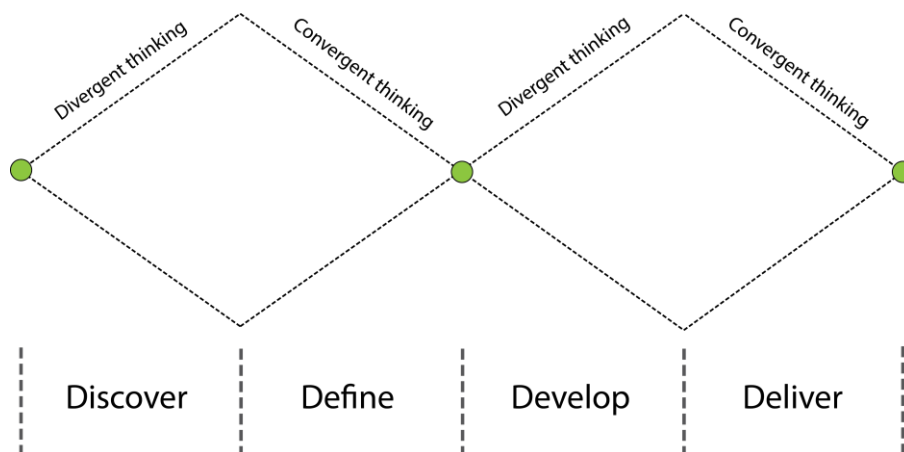


Figure 6 - Double Diamond, Adapted from Stickdorn et. al.

Discover

During the discovery phase as much information as possible about the customer, market and technologies is gathered i.e. through interviews, focus groups or observations.

Define

The information is then condensed during the defining phase through user analysis into a concept specification. The user analysis is usually converted into concept specifications.

Develop

In the developing phase concepts are generated based on the user demands and concept specification. This can be done in many different ways i.e. brainstorming, brainwriting or other creative methods.

Deliver

Lastly the concepts are reviewed and a winning concept is selected before moving on to the system level design. The selection is usually done through concept screening and scoring that rates the concepts depending on how well they fulfil the concept specifications (Ulrich & Eppinger, 2008).

2.2.2 Service Design

Service design is an interdisciplinary approach that combines different methods from various disciplines. The aim is to provide a holistic view on design and to make sure that the service provided is both useful, efficient, effective and desirable (Stickdorn & Schneider, 2013). There are many different approaches to describing the service design process. The point is to work in many iterations and to make improvements in every iteration. It is also important to work in iterations closely with the end user (IDEO, 2011). The use of iterations and close customer feedback makes sure that the development work stays on target throughout the whole process. Below is the service design process as described by Legeby (Legeby, Service Design - Powerpoint presentation, 2014) for Scania. IDEO's *Human Centred Design Toolkit* also covers the importance of working with the user in the beginning of the design process, that is where the real needs and problems will be found.

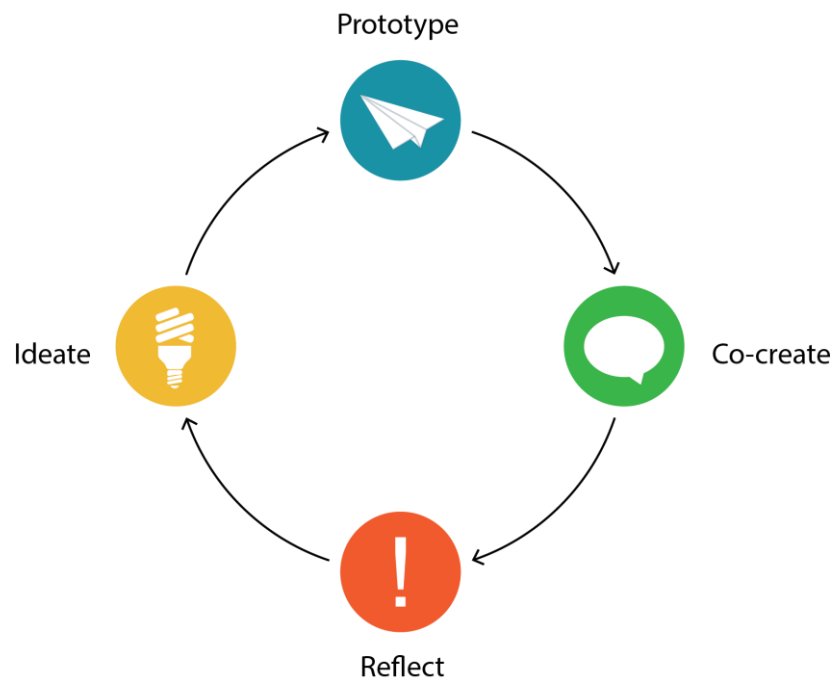


Figure 7 - Service Design Loop by Legeby, M.

The iterations in service design can be described as follows:



Co-create

The idea is to pick up as much information about the target user as possible in this diverging phase. That can be done through various user centred methods i.e. five whys, user observations or design scenarios (Stickdorn & Schneider, 2013).



Reflect

In this converging phase the information is analysed and mapped to identify the real user problems. The process could include user analysis, customer experience map and much more that clarifies the latent user needs (Legeby, Service Design - Powerpoint presentation, 2014).



Ideate

When a clear view of what the problems and needs might be solutions have to be created. This can be done with the help of different methods i.e. brainstorming, bodystorming or focus group workshops. The aim is to create as many relevant and innovative concepts as possible (Legeby, Service Design - Powerpoint presentation, 2014).



Prototype

The concepts are then realised through prototypes and mock-ups. By demonstrating these triggers to the customers, important feedback can be collected and new ideas

might spur from the testing and evaluation of the concepts. This information is stored and acts as new input into the service design loop. In this way improvements can be done and the following iteration will be better and have more depth to it. This process is iterated as many times as needed (Legeby, Service Design - Powerpoint presentation, 2014) (Stickdorn & Schneider, 2013).

2.2.3 Product Development and Service Design

The four initial phases in the product development process and the four phases in service design share many similarities. Service design however has a greater focus on finding what it is that brings value to the user whereas the product development process traditionally focuses on user needs and technical specifications. The product development process is most often a straight line from project initiation to product launch whilst the service design approach is more iterative and moves on when both developers and customers are satisfied. Both of these processes are guidelines more than manuals and may be adapted in order to suit the design team's specific needs (Ulrich & Eppinger, 2008) (Stickdorn & Schneider, 2013).

2.2.4 Personas

Personas are fictional profiles developed to represent a particular group based on their interests and habits. How good a persona is, is often shown in how engaging it is to the people using it (Stickdorn & Schneider, 2013). To make the persona more realistic and engaging it is important to add depth to the characters, that could be done with for example pictures of the character, more deep info like personal interests and hobbies, goals in life. (Nielsen L. , 2011). There should be some thought behind the use of personas and the method should be adapted to suit the desired result (Blomquist & Arvola, 2002).

2.2.5 Concept Generation Methods

To create good and valid concepts, some methods are needed and a set of methods that were used throughout this study are described here.

Brainstorming

Brainstorming is a method for generating ideas in groups or individually and is good for developing new concepts and solutions to a specific problem. The basic procedure is as follows:

- Selecting a group of three to ten participants with different backgrounds
- Posing a clear problem, question or topic to the group
- Let the group generate ideas without any attempt to limit the type and number of ideas. This is usually called the diverging phase and censorship is strictly forbidden and wild ideas should be encouraged
- Discuss and select ideas for further development. This step is often called the convergent phase of the brainstorming session (Wilson, 2013).

The simplicity of these rules propose that putting together a good brainstorming session is easy but in fact it is the opposite (Scanell & Mulvihill, 2012). Adrian Furnham states that that brainstorming has been proven to provide less innovative results compared to letting the participants generate ideas on their own. This is due to a number of factors e.g. social loafing which means that people tend to make less effort when in a group and there is someone else that can do most of the work. Another limiting factor is evaluation apprehension which suggests that people are afraid of telling their ideas with the risk of being mocked or made look stupid. The last factor is production blocking which means that since only one person at a time can suggest an idea the group is limited compared to everyone writing down ideas simultaneously.

Furnham however states that it is possible that “brainstorming groups fulfil other needs in the organisation which may or may not compensate for the resultant loss of creativity”. These factors could be an increase in decision acceptance, pooling of resources or to benefit from specialisation of labour (Furnham, *The Brainstorming Myth*, 2000). The opinions go apart and Gobble states that the best ideas emerge from sessions that offer enough freedom to explore without ranging too far from the question at hand. The key is to find a good balance that draws the best ideas from the group (Gobble, *The Persistence of Brainstorming*, 2014).

Brainwriting 6-3-5

This method is a variant on the traditional brainstorming method. Six participants have one piece of paper each. All the participants write down three ideas each on their piece of paper. When this is done the paper is passed on to the person sitting next to them. This is done five times. The initial idea with this method is just to use words but sketching can also be a variant. The positive and negative effect when using sketching can be that the sketches are misinterpreted. The benefit however could be that a person thinks of new ideas based on that sketch either interpreting it right or wrong (Linsey & Becker, *Effectiveness of Brainwriting Techniques: Comparing Nominal Groups to Real Teams*, 2011).

Random Words

The random words method is a quick and powerful method to come up with new and innovative ideas. The idea is to combine three different categories and use these as triggers for the new concepts. The categories are:

- A place
- A feeling
- An action

The session is initiated by letting the participants generate words in each of these categories for about a minute each. When all categories have been generated, three words, one from each category drawn at random and combined. The participants now have about two minutes to generate ideas that have to include all three words

in each of their ideas. The ideas are then presented and similar to brainstorming and brainwriting, criticism is strictly forbidden as even more ideas can come up from group discussion. When one session is done, another three words are drawn and more ideas are generated based on the new set of words. This process is repeated until the workshop leaders are satisfied. The Random Word method is frequently used at Scania and has been explained to the authors by the Driver Vehicle Interactions Department.

2.2.6 Concept Selection Methods

Concept selection can be done in many ways. One way is to discuss and consult experts for advice. There are other ways that show a more direct way of differencing concepts and one of them is described below.

Concept Scoring

Concept scoring is used when an increased resolution will help differentiate between concepts. The relative importance of the selection criteria is weighed and focus is spent on a more refined comparison with respect to each selection criterion. The scores are determined by the weighted sum of the ratings. Each selection criterion is given a score from one to five corresponding to its importance. The higher the score the more crucial that criterion is. When this is done each concept is rated based on how well it fulfils that specific criterion on a similar scale from one to five. The numbers are then multiplied and the weighted score ranges from one to twenty five. The scores are summarised and the concept with the highest score is the one that best meets the given criteria (Ulrich & Eppinger, 2008).

2.3 Data Gathering

When performing exploratory studies it is recommended to have some purpose to the data gathering. This is done to more easily find relevant data and not to collect too much irrelevant information (Yin, 2003) (Saunders, Lewis, & Thornhill, 2009). Some knowledge, preferably through literature studies, of the area of the study is necessary in order to be able to ask relevant questions during the user studies (Saunders, Lewis, & Thornhill, 2009).

2.3.1 Interviews

Interviews are one of the most important sources of case study information gathering (Yin, 2003). A user interview offer a flexible approach to gathering large amount of data in various topics like system usability, user experience, job analysis and many more. Depending on the data that needs to be collected the type of interview can be varied (Saunders, Lewis, & Thornhill, 2009). There are three major types of interviews; Structured interviews, Semi-structured interviews and unstructured interviews (Stanton, Salmon, Walker, Baber, & Jenkins, 2005).

Structured interviews goes through the same specified set of questions each time and should be asked in the same tone of voice in order to prevent bias (Saunders, Lewis, & Thornhill, 2009). Due to their rigid nature structured interviews are the least popular type of interview (Stanton, Salmon, Walker, Baber, & Jenkins, 2005).

Unstructured interviews do not have any pre-specified questions although the interviewer needs to have a general idea to what information he or she wants to find out. The unstructured interview is of a more exploratory nature but is not very often used due to the fact that crucial information might be lost or missed during the interviews (Stanton, Salmon, Walker, Baber, & Jenkins, 2005) (Saunders, Lewis, & Thornhill, 2009).

A semi-structured interview is somewhere in between an unstructured interview and a strictly structured questionnaire (Lindahl, 2005). By dividing the survey into themes rather than specific questions, more information could be gathered with follow-up questions when needed. The questions asked were open-ended and allowed the interviewees to elaborate their answers (Stanton, Salmon, Walker, Baber, & Jenkins, 2005). The follow-up questions could vary from one interview to the next (Saunders, Lewis, & Thornhill, 2009) and would generally be of a probing nature in order to get as much information out of that topic (Stanton, Salmon, Walker, Baber, & Jenkins, 2005). One important issue is to not influence the interviewee through body language or to answer the respondent by saying “yes” or “no” when the person talks.

2.3.2 Interviews Compared to Focus Groups

Interviews is a good data collection method since it can be conducted in the user's own environment. Griffin and Hauser also identifies that almost as many user needs can be identified through individual interviews compared to arranging focus groups (Griffin & Hauser, *The Voice of The Customer*, 1993). The correlation can be viewed in Figure 8.

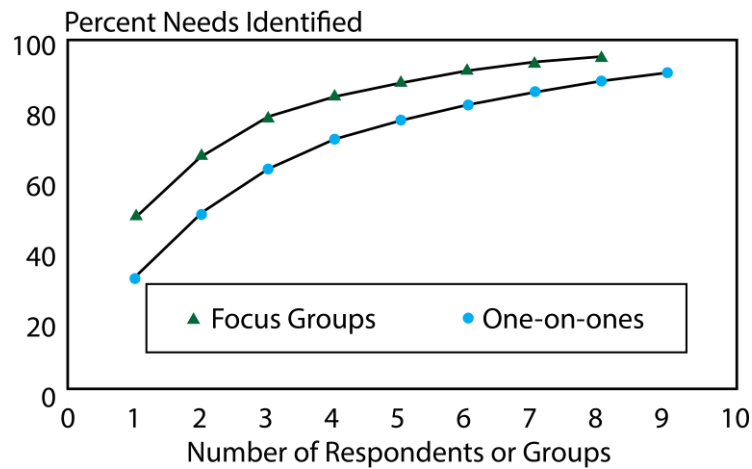


Figure 8 - Identified Customer Needs, Adapted from Griffin & Hauser 1993

2.3.3 Observations

Observations is a good way to further identify user problems by seeing the users in their real environment. Things that the users forget to mention or details that the user do not notice themselves can be identified. When performing an observation study there are a couple of different roles one can take. To get the most out a study the role as observer as participant can be used. The role of observer as participant is like attending an activity to observe without taking part in the actual activity, in other words one would be a spectator. However, the identity of the participant as a researcher should be clear to all concerned. The advantage is that one can focus more on the researcher role. Focus is spent on the discussions with the participants. Much of the research work as an observer or a participant relies on the building of relationships with others. The researchers need to be flexible in their own personality to some extent, one's own personality must be suppressed to a greater extent and that is something not everybody feels comfortable with (Saunders, Lewis, & Thornhill, 2009).

2.3.4 User Evaluation Methods

In order to extract as much valid data out of the user testing sessions, a number of user evaluation methods have been gathered. These methods are described below

Expectation Measures

Expectation measures is a way of measuring the experienced difficulty of a specific task. This is done by asking the test subject how difficult they expect the given task to be based on hearing the task instructions. The test subject is asked again directly after the task has been performed on how difficult they actually experienced the task to be. The task difficulty is measured in seven increments ranging from 1 meaning (very difficult) to 7 (very easy). The difference in values before and after is then displayed in a scatter plot to visualise the concept performance.

There are four different areas that can be identified in the plotted area. The first are tasks that appeared easy but in fact were more difficult. These are concepts that should be “fixed fast”. Secondly comes the tasks that appeared easy and also were easy. These concepts correspond to the user’s expectations and are labelled “don’t touch”. Third comes the concepts that appeared difficult beforehand but turned out to be easy. These concepts fall under the “promote it” label and are a pleasant surprise for the users. Fourth and last comes the tasks that appeared difficult and also were difficult. These are labelled a “Big Opportunity” since they have great potential to see some improvements. The different areas can be viewed in Figure 9 below (Tullis & Albert, 2013).

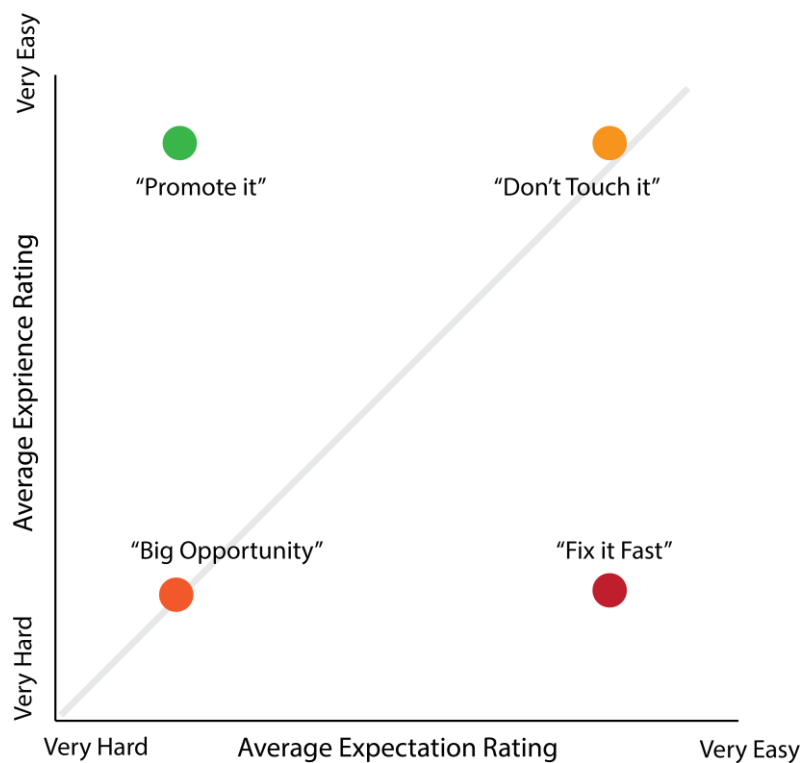


Figure 9 - Expectation Measures, Adapted from Tullis & Albert 2013

System Usability Scale

The system usability scale was originally developed by Brooke in 1996 as a quick and dirty survey that would allow the usability practitioner to easily assess the usability of a given product or service. The SUS instrument is composed of ten questions on a five point scale of strength of agreement ranging from fully disagree to fully agree. The questions alternate between positive and negative and care must be taken when calculating the final score. The SUS-score has the lowest score 0 and the maximum score 100 and is calculated as follows:

$$\text{Total score} = ((Q1 - 1) + (5 - Q2) + (Q3 - 1) + (5 - Q4) + (Q5 - 1) + (5 - Q6) + (Q7 - 1) + (5 - Q8) + (Q9 - 1) + (5 - Q10)) * 2,5$$

Equation 1 – System usability scale calculation

The Q stands for the result in the specific question(1-5). This method gives a good reference on how good the usability of a product is and the value can be communicated between experts and non experts easily (Bangor, Kortum, & Miller, An Empirical Evaluation of the System Usability Scale, 2008).

3 Method and Implementation

3.1 Process

The development process that was chosen for this project is a combination of the product development process described by Ulrich and Eppinger and the service design toolkit described by Stickdorn et al. The idea is to test the user centred and iterative workflow of the service design toolkit while still relying on the well tested product development process. The combination will hopefully create a good synergy effect and result.

This is a flowchart of the work process. To easier show how work was conducted and how the different stages in the process has gone hand in hand with each other this illustration was made. The different sizes of the circles represent how comprehensive each stage has been. The colours of the circles origins from the Service Design Loop by Legeby, M this is to try to clarify what was done and when it was done.

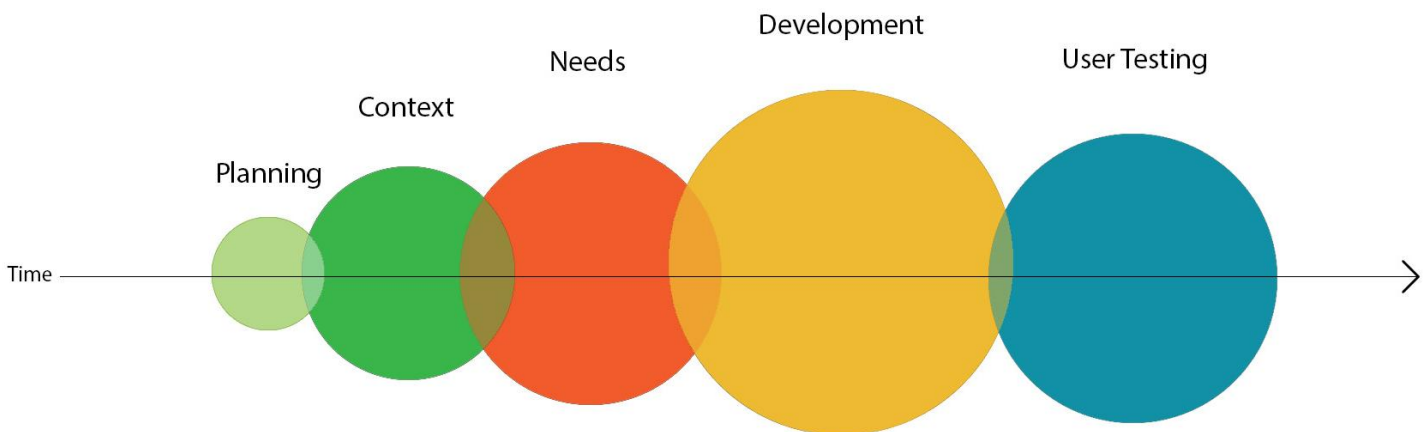


Figure 10 - Project Workflow

3.2 Planning

At the start of the project a rough plan was made in order to get a better overview of the tasks and gates that needed to be completed before moving on to the next part of the project. The different tasks were identified and written down in a work breakdown structure (WBS). This was later used when the tasks were placed in order and given a specific time for when they needed to be completed.

3.2.1 Gantt Scheme

In order to get a good overview of the work and all its different parts a Gantt scheme was created. By having a clear and consistent plan, resources and activities can more easily be managed and the potential outcome of the project is improved (Tonnquist, 2010).

3.2.2 Near Zone Planning

It is often hard to make a detailed plan when tasks are far off into the future. A good thing to do then is to plan the near future and only make a rough plan of tasks further ahead. The plan is then later revised as more information is gathered and more knowledge is gained (Tonnquist, 2010). At Scania this type of planning is well implemented through the visual planning method (VP). VP is done in periods of five weeks and each person is responsible for their own planning. By using different coloured sticky notes for each type of activity a good overview of the workload can be achieved. This overview is even better understood when all the co-workers at the group put their time plans next to each other. The overall work intensity can easily be seen and work can quickly be redistributed if needed to another resource (Scania AB, 2013).

3.3 Context

Those in control of the details of a product must interact with customers and experience the user environment of the product (Ulrich & Eppinger, 2008) and to create innovative and well working ideas one needs to know their end user well (Stickdorn & Schneider, 2013). This chapter describes the method used in this project to map the problem, understand the target user. The method is described chronologically and will hopefully provide the reader with a good understanding of the identified user problems.

3.3.1 Literature Studies

Decision-making in this project needs to be based on facts and to incorporate this, relevant literature is needed. The literature has been gathered through extensive research via academic databases, textbooks on relevant topics and through presentations given by experts in their respective fields.

3.3.2 Benchmarking

To prevent “reinventing the wheel”, research was made to investigate what has been done before in the field of gaze interaction as well as adjacent fields. The benchmarking was done through academic journals, expert presentations and videos from leading manufacturers demonstrating their technology and its applications.

3.3.3 Meeting With Tobii

Tobii technology is the world leader in gaze and eye tracking technologies (Tobii Technology, 2013). To get a better understanding of the potential uses of the gaze technology, a meeting with Tobii was held at Scania. Tobii demonstrated how the eye tracking device works and its technical potential and limitations. It was also well demonstrated how limited the human vision is and that it is more important to observe where the eye is not looking rather than where the user has its focus. The sales representatives meant that this is the most important issue when developing safe and well designed driver systems. A visit was made to the driving simulator and a discussion was held on where to place the eye tracking device and how wide an angle it would pick up of the driver’s field of view. The device might not be able to pick up the entire width of the field of view i.e. if the driver is looking in the mirrors. This is dependent on where the device is placed and what one wants to measure. If the entire field of view is to be measured then more than one device is needed.

Performance in Changing Light

The light varies a lot and rapidly when driving in a vehicle on the road. A technician from Tobii ensured that the accuracy of the eye tracking technology would be within the desired range for the purpose of the study.

3.3.4 User Study

In order to get a deeper and better understanding of the target group, two field studies were conducted. It is good to select a user group that is representative of the main user group in order to get a relevant result (Stanton, Salmon, Walker, Baber, & Jenkins, 2005). Therefore the target groups were chosen both from short distance distribution trucks and long haul truck drivers as these might have a different set of problems that need solving. Through literature research and meetings with the statistical expert in the Driver Vehicle Interaction Department, a user study was put together. The focus of the study was to get a good view of the drivers’ everyday and to identify potential problems that could be solved through the gaze interaction technology. The study aimed to highlight every type of problem, from annoying details to actual hazards in the driver’s environment.

Study of Distribution Truck Drivers

The study was conducted at a small company situated in Västerås and their main job is to deliver milk and other groceries to local stores in the area. Two distribution truck drivers were studied during one whole workday in separate vehicles. Interviews were conducted in parallel with observations in order to identify

problem areas. A typical workday is nine hours long and consists of many stops where the driver has to park, unload goods and load empty pallets before driving to the next stop. The job is high paced and often under time pressure. A distribution business is different from long haulage driving since the driver has to perform many different functions apart from just driving. Many of the drivers often experience stress since they often cannot plan their own workday but instead have to follow a tight time schedule. Small improvements in the driver environment make a big difference if used frequently.

The study can be found Appendix in 8.1

Study of Long Haul Drivers

The target group was studied during a four day trip from Södertälje, Sweden to Zwolle, Netherlands and back. The study was done with the Transport Laboratory which is a transport company connected to Scania. By studying the drivers in their working environment for an extended time period, a better understanding of how different factors i.e. fatigue, daylight and weather affect the drivers' workload. The study was conducted with thirteen drivers and the main focus was for the authors to get a good understanding of what driving a truck for a long period of time feels like and not to get statistical data of usage.

Each driver drives the truck approximately four and a half hours before having a break. The truck is then given to another driver that continues to drive south whilst the first driver receives a truck that is going north. This system allows the trucks to move continuously between Södertälje and Zwolle while the drivers drive their specific route back and forth. The system also takes into account the drive time limitations and break times that are necessary by law. By following a specific truck the study could include up to four drivers per day. However, some drivers preferred to drive the whole way and stick with the same truck throughout their working day which meant fewer respondents. The first two days the study was conducted during daytime. On the way back from Zwolle to Södertälje however, the entire way was covered in one stretch and studies were conducted with the night time drivers as well. Driving during both night and day meant a better view of the different driving scenarios a driver may face during their work day. The study can be found in Appendix 8.1

Interviews

In this particular study, the focus was to understand the drivers and the challenges of their working environment, semi-structured interviews were used with a probing nature. Prior to the interview the drivers were presented with a brief introduction and background of the study but not too much information as this also might affect the respondent's answers (Lindahl, 2005). The interview was finished with one final question; if the driver had something to add to the study or if something was missing. After the interview was done, a full and more in depth explanation was given to the driver if it was requested (Lindahl, 2005).

Observations

Being subject to an interview can be intrusive (Saunders, Lewis, & Thornhill, 2009). In order to get as much reliable data about the users' thoughts and actions the drivers were first asked to perform a certain task and then to repeat it while explaining the different actions involved that specific task. This was done in order to see if there was any difference in the drivers' action compared to what they said they were doing. The observations were, when possible, focused on some major areas i.e. when the drivers had to look away from the road or when the drivers had to take their hands off the steering wheel. This was done to more easily identify problem areas that most likely could be helped through the use of gaze interaction technology. The observations were carried out in parallel with the interview and special attention was given to the situations when the drivers answered one thing but did another.

3.4Needs

To fully understand all the input data and to summarise it into a coherent problem description an analysis needed to be done. When all the different data was gathered, it was summarised and resulted in the final user needs. This chapter describes the different sets of data that were analysed.

Analysis of the User Interviews and Observations

After the user studies were done, the data and comments from the user study were reviewed and summarised into a single document together with observations relevant to where the drivers' attention were directed or any problems that the drivers might experience during their work day. The observations also documented what actions that forced the driver to look away from the road while driving. Any comments about the user interface, the information in it or functionality that concerned driving was also written down. Comments that were not relevant to the area of interest (i.e. a special compartment for snus or the installation of an in-truck toilet) did not get written down in the summarising document. To get a rough overview on how common a problem was among the drivers, each statement was counted and the frequency of which it occurred was written down next to the statement. The summarising document can be found in appendix 8.2 and the results from the study can be found under heading 4.1.1

Information from MODAS

Through a research project conducted by Scania, Uppsala University, Interactive Institute in Piteå and Luleå University of Technology additional user information has been gathered. The research project is called Methods for Designing Autonomous Systems (MODAS) and investigates the development method for interfaces in future autonomous trucks and the potential opportunities and demands that this will require from the driver. Throughout this project, extensive user studies have been conducted and a number of needs are identified based on a weighted score. The score is based on user interviews where the users rated driving situations combined

with the frequency of their occurrence. The MODAS user study provided similar conclusions to the long haul user study conducted by the authors. This gives credibility to the identified user needs and increases the likelihood of solving actual user problems. The top rated user needs can be found in appendix 8.3

Summarised User Needs

The data from the user studies, the MODAS project and expert advice from the developer team at Driver Vehicle Interaction Department was all analysed and combined into specific design problems. These problems were formulated in order for the continuing development work to solve an actual problem and not satisfy a need that does not exist in the target user group. The problem formulations and observations can be found in user study results 4.1

Concept Specifications

The summarised design problems resulted in the creation of the concept specifications. The concept specifications are requirements that the concept needs to fulfil in order to satisfy the customer needs (Ulrich & Eppinger, 2008). The concept specifications can be found in the result section.

3.5 Development

3.5.1 Concept generation

This chapter describes the concept generation phase and the idea generating workshops that were held and description of the participants.

Workshop 1

The first workshop session in this project was a workshop that was customised to suit the desired outcome. The workshop is described below.

Workshop Method

The workshop had a good participant size (Scanell & Mulvihill, 2012) of nine and started with dividing the participants into small groups of two to three people and each group was given a persona to have an extra input when to generate ideas.

To get a better and more realistic workshop with solutions that satisfies an actual need, personas were used. For this specific workshop, personas previously developed by Scania were used as these are well worked through and are based on user data and values that the authors do not have access to. In this way the workshop could result in more accurate results and a deeper user understanding by the workshop participants.

The idea was that each group would generate ideas specifically for their own persona. To their aid each group had pens, sticky notes and the persona description in front of them with both text and picture. The groups were then presented with the different scenarios, one at a time.

With the identified problems and needs from the user study conducted with the drivers from the Transport Laboratory, scenarios had been built around these problems (Appendix 8.4.1). Each of the scenarios covered one or more specific user needs that the user study had identified. The needs used to form the scenarios were the problems that occurred most frequently among the studied subjects. The identified needs were matched with needs that had been identified earlier in pre-studies for the MODAS project. Each group were given the same design scenario as the others and were given five minutes per scenario to generate as many ideas that they could.

The method of brainstorming was used and for each scenario the participants would write down their ideas on a sticky note or a paper. The participants would start off by writing down their own ideas silently.

After each scenario of five minutes the participants all had to present their ideas, after which a small discussion took place. This was done with all the scenarios, four in total. Some scenarios generated many ideas while others generated less. The workshop lasted for about an hour and at the end of the workshop, all the ideas were collected and ordered according to which scenario and persona it belonged to.

Workshop 2

To get even more ideas for concepts a second workshop was held. This one was more focused on the technique and functions and the aim was to get more ideas and more innovative and specific concepts. A large number of ideas improves the likelihood of finding a great idea. The number of participants in this workshop was six and there were no groups formed, all of the participants worked as a big group instead.

Workshop Method

The workshop was divided into two parts, one were the passive aspects of the technique was examined and one were the active functions of the technique was looked into. The core methods used in both parts were brainstorming, brainwriting and random words. Some minor adjustments were done to these methods to adapt

them better for the situation. To adjust your methods for the situations is a key thing when designing, and as a designer you need to be able to use your design toolkit in different ways depending on the situation (Ulrich & Eppinger, 2008) (Blomquist & Arvola, 2002).

Passive Gaze

It started off with a short info about how the technique can measure almost everything the eye can do, to inspire the participants. The participants used a form of brainwriting for generating ideas connecting the info about the eye and the eyetracking technique.

Each participant generated three ideas each on a piece of paper during three minutes. After the three minute session the participant sent their paper to the next participant, so that everybody got a paper with new ideas. Next step for the participants was to continue working on the existing ideas and this was also a three minute session. A short presentation with the ideas was held after each round. After that session the procedure was repeated until the papers were full and by that time three rounds had passed. The participants were set to pick one of their ideas they liked or believed in, then they took that idea and passed it to the person next to them while they received another idea from another person. The received idea would work as input for the further development of the ideas. This was just done one time and the participants had three minutes here as well.

A brief presentation from each participant was held just so everyone could hear and understand the ideas and get a chance to reflect on the other participants work. Afterwards open discussions with all the participants were held where everyone could discuss the different ideas and add critique or new inputs to the ideas.

Active Gaze

The participants in the workshop used brainstorming to generate different triggers, functions and feelings. A trigger in this scenario is an action used combined with the gaze technique to deal with the Midas touch issue i.e. buttons, voice etc. Functions were things that the driver wanted to happen or things to get activated; it could be everything from changing radio station to getting the ETA (estimated time of arrival). The last category was feelings which basically was just feelings and emotions that the participants could come up with.

Two minutes were appointed for each category (triggers, functions and feelings) to generate the words on a post-it. The generated words were put in three separate piles. Then one post-it from each category was randomly selected, one trigger, one function and one feeling. The random selected words were used for generating and creating new concepts and ideas. Every round with the three random words was two minutes. This was repeated three times and after every round the participants presented their ideas they had come up with.

Workshop Participants

The participants of the first and the second workshop were people from the department. The people working on this department are mostly vehicle ergonomists, human factors engineers, sound designers, interaction designers and industrial designers. The participants therefore had a good insight in the subject and are confronted with problems concerning vehicle interaction almost every day. They were well aware of the gaze interaction technique and what it could do and not do. The participants' knowledge in the technology and its limitations might have had a negative impact on the creative outcome of the ideas.

3.5.2 Concept Selection

When all the concepts were generated the concept selection took place to limit the number of ideas. Not all the concepts would have the possibility to be tested in the simulator and the choices and decision are described in this chapter.

All the generated concepts were discussed and analysed. The concepts that were very similar to each other were combined into one concept. Feedback was received from professional test drivers at Scania and expert reviews were made by the people within the department.

Level of Gaze Implementation

The concepts were sorted into a chart showing if a concept is active or passive and to what level the gaze interaction is appearing in the concept (redundant and dominant). The chart showing the final concepts can be found in chapter 4.3.1.

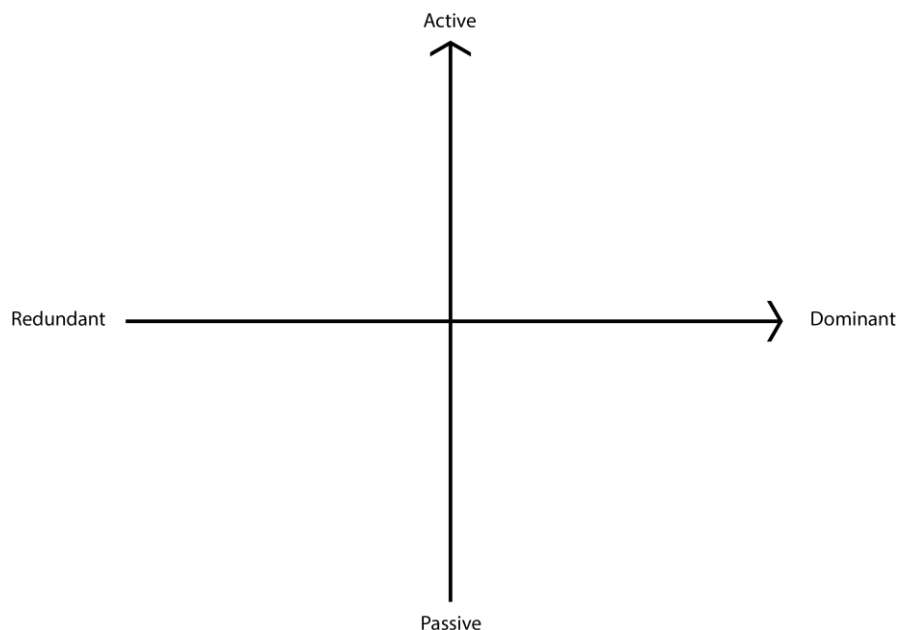


Figure 11 - Level of Gaze Implementation

The chart shown in Figure 11 illustrates if a concept is passive or active which means if the concept uses active interaction triggered by the user or if the concept works without the user even noticing it. The other axle indicates if the concept controls a function that is redundant and therefore can be navigated in many different ways or if its functionality is strictly based on the gaze interaction functionality and cannot be activated in any other way. The point of the chart is that it makes it a lot easier to classify the generated concepts. By positioning the concepts on this chart, it can quite easily be assessed if a concept brings novelty and is worth proceeding with or not. The aim of this project is to evaluate the gaze interaction technology and the concepts tested will preferably come from the right part of the chart and both from the active and passive sections.

Concept Scoring

Based on the concept specifications that were created previously, a session of discussing and selecting concepts began. Concepts that in some way did not fulfil one or more of the concept specifications were removed until only five final concepts remained.

These final five concepts were all potential winners and a scoring session took place. Based on how well a concept fulfilled a certain criteria it would receive a score from one (not good) to five (very good). The weighted scores were calculated and the result can be found in chapter four.

Head Up Display (HUD)

The human eye is quick on detecting objects that move in the peripheral field of view according to Johnson (Johnson, 2010) and Tobii technology (Tobii Technology, 2010). The idea is that the more time the drivers look at the road, the likelihood of a crash or near crash event decreases. This correlates to the NHTSA guidelines for not looking away from the road for more than 2,0 seconds (National Highway Traffic Safety Administration, 2012). The head up display technology has a great advantage of projecting the desired information closer to the forward road scene and thus keeping the traffic in closer proximity to the human peripheral field of view. Applying the information on how the human vision works according to Johnson means that a head up display could increase the time spent looking on the road and decreasing the likelihood of dangerous situations. The authors chose to move forward with this information and to incorporate it into the concept development process.

Technological Feasibility

Since the aim of this thesis is to investigate the potential and setbacks of the gaze interaction technology, only concepts that could show both these sides were chosen. The testing needed to be done on a more general level rather than testing a specific function.

Implementability in Simulator

Another factor that played part in the selection of the final concepts is the implementability in the simulator. The final five concepts were discussed with the simulator system development expert and he could quickly give an answer to which concepts that were possible or easier than others. A few of the concepts required instruments that simply were not possible to get for the simulator testing such as physical objects that would react to the gaze data, a task that time did not allow.

Final Selection

With all the concept specifications, limitations in the simulator and aims of the thesis added up, only two concepts remained.

Further improvements

With the final concepts chosen some minor tweaks and adjustments were done in order to ensure that the concepts would work in the simulator. Improvements were also done by discussing the concepts with some of the Scania test drivers.. The Scania test drivers have a lot of experience with reviewing new concepts and providing their input.

3.6 User Testing

To make sure that the generated concepts actually work and also to assess whether the gaze interaction technology is suitable for the truck driving environment the technology needed to be tested in a driving environment. Due to the budget of the project and the need for strict repeatability all the concept testing took place in the Scania driving simulator.

3.6.1 Simulator Setup

The simulator at Scania consists of a number of projectors and computers in a local network. In the centre of the room stands a truck interior featuring steering wheel with fully functioning buttons and an air pressured chair to mimic the real driving experience. Three big screens represent the forward and side road scene and an additional four screens are paired two each to represent the rear view mirrors. To enable the functionality of a head up display an external projector was used to project additional information on the main screen. This was done because of how the computer that run the main programme is configured. The difference in screen resolution also enhanced the illusion of having the information projected into the air in front of the driver. The simulator also features a fully functioning instrument cluster that can be navigated with the buttons placed on the steering wheel. This corresponds exactly to how the instrument cluster works in a normal truck that exist on the market today. Therefore most professional drivers have a good understanding on how this system works. To increase the level of realism and to give the driver additional feedback the simulator also features a sound system. The system responds to the level of acceleration that is given in the truck and the driver can therefore more easily hear if he or she is accelerating or decelerating.

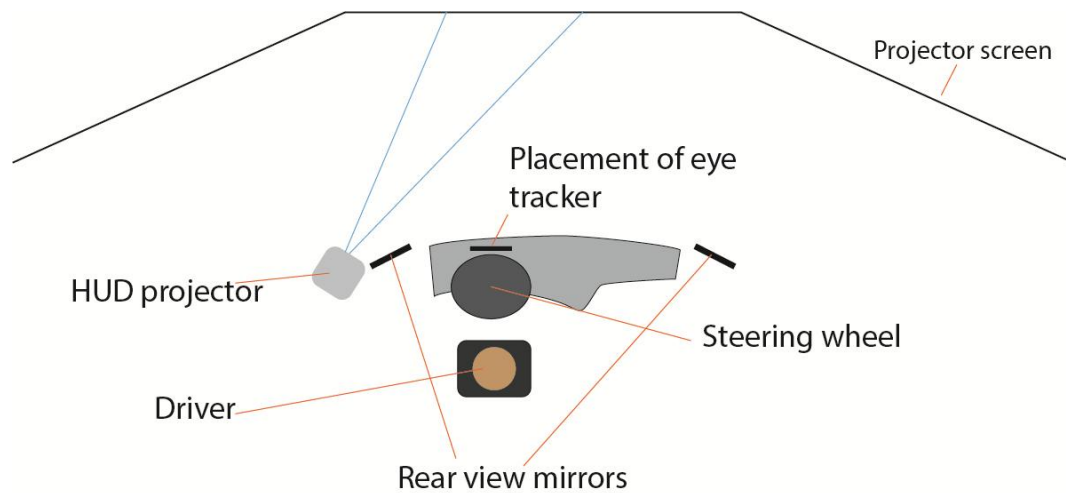


Figure 12 - Simulator Setup

The simulator computers run a simulation software that is a joint venture between a number of Swedish companies and the Swedish National Road and Transport Research Institute (VTI). In the software it is possible to design specific driving maps, scenarios and tasks. Events can be triggered and obstacles can be placed in order to distract or help the test participant. There is also a number of different types of data that can be extracted from the simulator. This data can be logged every fifty milliseconds and stored into log files for later reviewing. The types of data can be viewed in chapter 3.6.6.

3.6.2 Eyetracker Setup

Since the eyetracker has a few limitations regarding the traceable area of viewing, the main focus of the study was to measure the gaze upon the road. Therefore the tracker was placed just above the instrument cluster. In this way the traceable field of view was maximised and most of the forward road scene could be captured and traced. When setting up the eyetracker, its position relative to the plane it is measuring gaze data on, has to be defined. This is done through measuring the distance (X,Y,Z) and angle to the centre of the defined plane. For this study the eyetracker was defined to correspond to the main screen in front of the driver. The main screen sizes were measured and then later the distance to the eyetracker. All the measurements were typed into the eyetracker's calibration file and only needed to be done once. When recording the gaze data it was also possible to set up a secondary camera to film the user. This possibility was used but the camera was directed towards the road and instrument cluster. In this configuration most of the task completion times could more easily be monitored. Another good thing about filming the road as the driver was operating the vehicle is that the concepts that used the secondary projector could be seen afterwards in the user camera view. This would not have been the case with a simple screen recording,

3.6.3 Head up projector setup

The gaze tracking system was calibrated to work with the main screen in front of the driver and for the head up projector to be working properly it had to be calibrated to the same plane. This was done by defining what coordinates that the projector screen occupied in the larger screen. More precisely this was done by showing a smaller calibration area on the main screen and warping the projector screen to match.

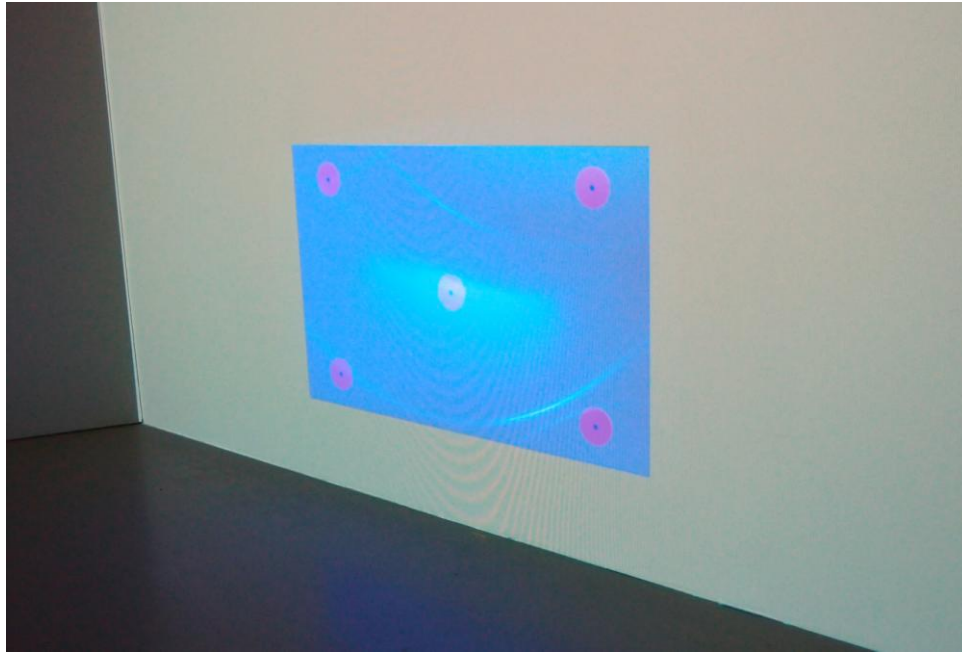


Figure 13 - Calibrating projector screen

When this was done the proportions of the small screen relative to the big one were calculated using measuring tape and a calculator. These proportions were then added into the software specially developed for this study by the simulator system development expert. With the software compensating for the new screen coordinates based on the reviewed gaze data the projector screen could now also work as an interactive surface.

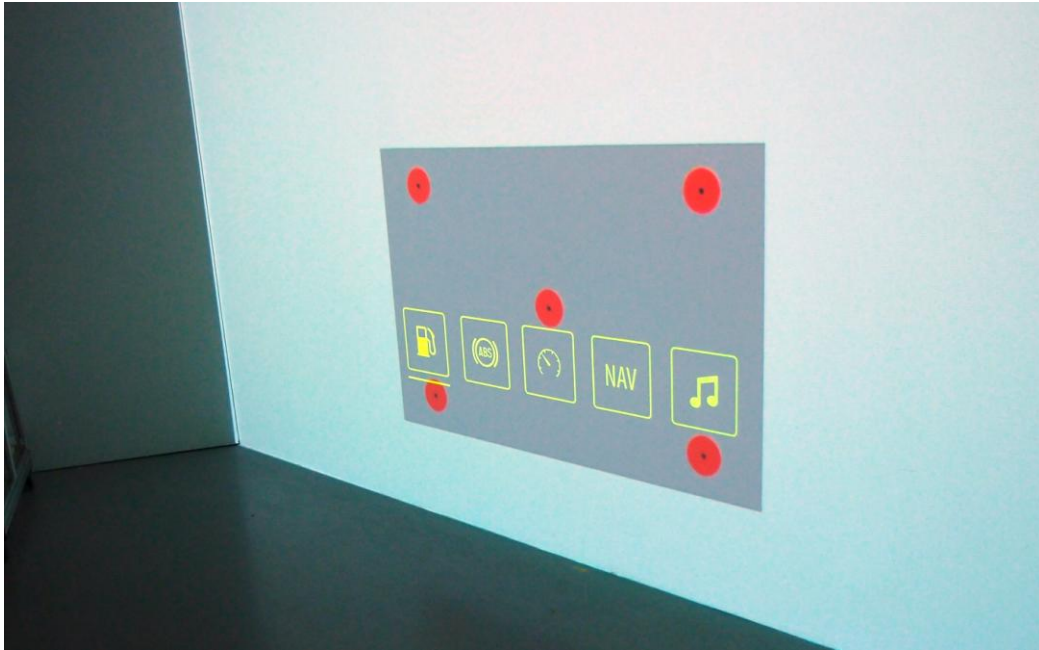


Figure 14 - Projected Interactive Surface

The interactive concepts that were developed for this study were implemented in a flash program that acted based on the gaze data that was fed to it by the eyetracker. Using this rather complex setup enabled the study to both have an interactive system and at the same time record the user gaze data.

3.6.4 Area of interest

To further determine if the driver's gaze was in the right area of the forward road scene, a virtual box was drawn out using a screenshot from the simulator. The area of interest was drawn out as an estimation of where a driver should look to have an increased likelihood of observing what is going on in the traffic ahead. By looking at the screenshot of the road and then observing where the peripheral field of view ended roughly, the area of interest was defined using post it notes placed on the projector wall.



Figure 15 - Area of interest forward road scene

When the area was defined the coordinates were identified and measured with measuring tape. They were then inserted into the software development kit and a polygon was created in the gaze interaction software developed by the simulator system development expert.

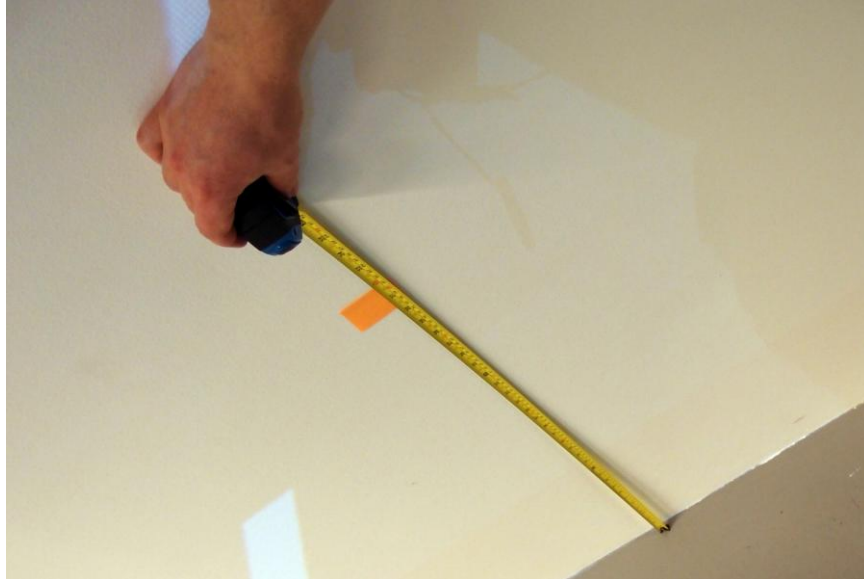


Figure 16 - Measuring screen coordinates

3.6.5 Setting up the test and pilot testing

When the eyetracker had been placed in a satisfactory position and the head up display projector was calibrated to correspond to the right coordinates the designed

scenarios could begin to be tested. The first testing sessions consisted of verifying that the technology was working and showing the right data in the right place. This was an iterative process and contained many small adjustments to compensate for any errors in calculations or replacing bad cables. When the technology was working properly the testing moved on to pilot testing of the driving scenarios. During this procedure, simple errors in the test structure and explanations were identified and corrected. The pilot testing was conducted with the authors, experts from the Driver Vehicle Interaction Department and other people familiar with simulator testing and test procedures. When the pilot tests were completed the test procedures were refined and adjusted to better suit the aims of the simulator testing.

3.6.6 Data gathering

The driving scenarios were constructed based on the type of data that could be extracted out of the simulator setup. All the recorded data was saved into a log file with a timestamp every fifty milliseconds. The data is listed as follows:

- Simulator timestamp
- Distance travelled by the truck
- Steering wheel angle
- Brake pressure
- Vehicle speed
- Event ID
- State ID within the event
- Event local timestamp
- Which button that is pressed on the steering wheel
- Validity of left eye
- Validity of right eye
- 3D coordinates of left eye
- 3D coordinates of right eye
- 2D coordinates of left gaze projection
- 2Dcoordinates of right gaze projection
- Left eye pupil diameter
- Right eye pupil diameter
- X coordinate within smaller projected screen
- Y coordinate within smaller projected screen
- Distance to the truck in front
- Warning type (fixation or tracking lost)
- Warning level (no warning, level 1, level 2)

To further increase the depth of the study material the task completion time for the secondary tasks was monitored with the user camera that was mounted next to the participants' right shoulder. Throughout the entire test, user comments about the concepts, test setup or simulator were written down to try to capture the user attitudes and potential improvements.

Expectation measures

Before each driving condition the participants were asked to rate how difficult a task would be and again to rate it after the driving condition was completed in accordance with Tullis & Albert (Tullis & Albert, 2013). The original scale ranges from 1 to 7. The authors however wanted a finer granularity and chose a scale from 1 to 10 where 1 = “very easy” and 10 = “very difficult”. The results from the expectation measures test can be viewed under the results section.

System Usability Scale

After each participant finished driving the last condition in each of the two driving sessions, they were asked to fill out a System Usability Scale questionnaire as described by Brooks in Bangor et al (Bangor, Kortum, & Miller, An Empirical Evaluation of the System Usability Scale, 2008). The questionnaires only regarded the concepts that were being tested and not the baseline and secondary task based conditions.

3.6.7 Test procedure

All the driving scenarios consisted of the participant driving a truck on a typical Swedish highway. In front of the participant was another truck that varied its own speed between 75-85km/h. The rest of the traffic was generated at random to create a realistic surrounding. The road was a two lane highway and the participant was told to only stay in the rightmost lane. All the secondary tasks were given in Swedish and all of the participants except one speaks Swedish natively. That participant got all the tasks explained to him as they appeared. Again the expectation measures were recorded and the driver could give general comments at any time during the driving session.

The order of the conditions was as follows:

- D - Baseline
- E – Secondary tasks
- F – Gaze support
- A - Baseline
- B – Secondary tasks instrument cluster
- C – Head up display interaction

This was done because the head up display being the most active concept. It was decided that this concept should be tested last to allow the drivers, if not used to the simulator, to warm up to the task. All the participants had the same order of the conditions but as previously stated a unique order of secondary tasks. The order of conditions was chosen, in collaboration with the simulator system development expert and mentor at Scania, as the most logical way of presenting the driving scenarios for the participants.

Concept – Head Up Display Interaction

This scenario consisted of three driving conditions, all taking about five minutes each to complete. The data that was measured in the three scenarios was the same for later comparison. The individual scenarios are described below:

Condition A – Baseline

The participants were given the task of keeping an even distance to the truck in front. Before the session started they were asked to rate the difficulty of the task at hand. Afterwards the participant were asked the same question on task difficulty as well as if they had any general comments regarding the driving condition and how they experienced the test.

Condition B – Instrument Cluster (ICL)

In the second condition the participants were asked again to keep an even distance to the truck in front but with four secondary tasks. The secondary tasks were the same for every participant but their order was rearranged in a way that no participant would receive their tasks in the same order. The secondary tasks were the following:

- Find average fuel consumption
- Display the total driving time
- Show the carriage weight
- Show the phone list (last made calls)

The secondary tasks were all located in the instrument cluster and the navigation process could be reviewed in the user camera afterwards.

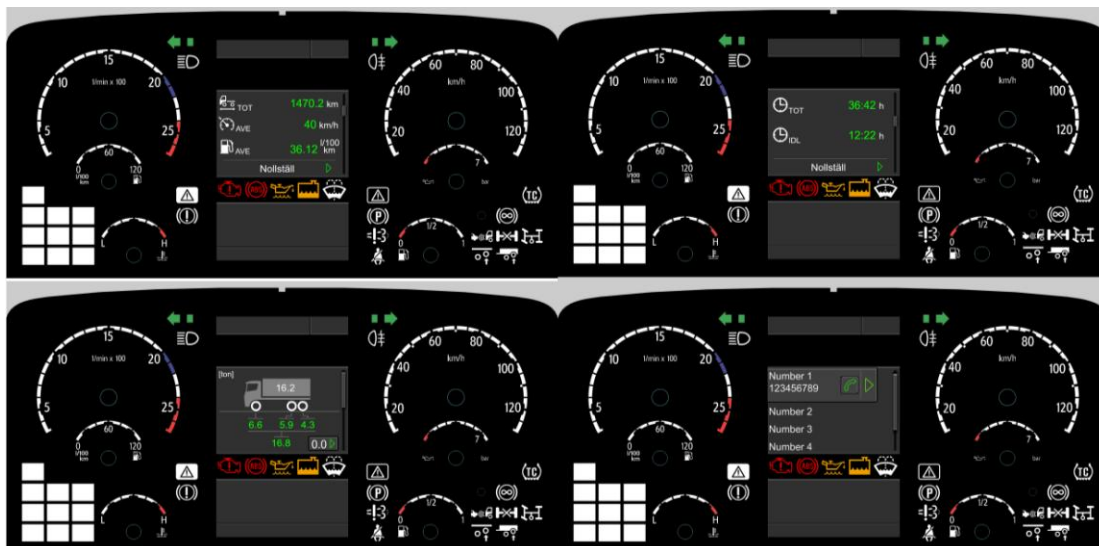


Figure 17 – The secondary tasks in condition B

Condition C – Head Up Display Interaction (HUDI)

The last condition was initiated by a quick explanation on how gaze could be used as a way of interacting and controlling the information around the driver. The participant was given a course in how the interaction was done with the head up display system. The participant was then again asked to rate the expected difficulty when they understood how the gaze interaction system worked. Four secondary tasks were given. Similar to the previous scenario the four secondary tasks were the same for every participant but were again rearranged in a way that no participant would get the same order of tasks. Due to patent application the secondary tasks cannot be displayed here. The tasks were arranged as follows:

- Display speed
- Show the average fuel consumption
- Show navigation
- Show music

The participant was again asked to keep an even distance to the truck in front and to drive the truck as if they were driving on a regular highway. Afterwards the expectation measure was taken again.

Concept – Gaze supported driving

Similar to the other scenario this scenario also consisted of three driving conditions, each taking about six minutes to complete. Once again was the same data measured in all three scenarios for later comparison. The scenarios took place as follows:

Condition D - Baseline

The participant was asked to keep an even distance to the truck in front even though the truck varied its speed. Expectation measures were taken and no secondary tasks were given.

Condition E – Secondary Tasks Without Support

Prior to the beginning of the scenario the participants were given a cell phone and were instructed on how the phone was unlocked. The participant was again asked to keep an even distance to the truck in front but had to complete four secondary tasks namely:

- Raise the temperature 8 degrees
- Call Kalle Andersson of the cell phone
- Display the driver points
- Turn on the hazard lights

No more information than that was given and the participant would then navigate the phone's menu by themselves. The order of the secondary tasks was rearranged for every participant to avoid learning of the system.

Condition F - Secondary Tasks With Gaze Support System

For the final condition the gaze support concept was tested. The drivers had the same task of keeping an even distance to the truck in front and at the same time completing the following four tasks:

- Turn the fan to number four
- Show the differential lock
- Call Simon Katzman on your cell phone
- Show the driving tips

The tasks also went through the process of being rearranged for every unique driver. Whenever the participant would look away for more than 2,0 seconds from the predefined area of interest, in accordance with the NHTSA recommendations (National Highway Traffic Safety Administration, NHTSA, 2012), the participant would get a warning triangle to light up and flash in front of the driver. If the system did not register gaze upon the forward road scene within another 2,0 seconds the warning level would increase and a signal would be added to alert the participant of their inappropriate behaviour. The system also alerts the participant if a fixation is registered for more than 2,0 seconds, this time however with a blue triangle. If the fixation does not stop the level would increase and the same warning sound plays to alert the participant.

3.6.8 Test Analysis

When all the testing sessions were done the work with analysing all the gathered data began. This chapter describes the result from the different variables that were incorporated into the analysis.

Expectation Measures and System Usability Scale

The data from the expectation measures and the system usability scale was transferred into two separate excel documents. The results can be viewed in chapter 0.

Lane Placement

One parameter that was interesting was the lane placement. The video and log file data was analysed and each of the tasks from the B and C conditions was logged in duration. The most interesting part of the lane placement data was when the participants were doing a task, as this is when the driver is most prone to lose stability on the road. The standard deviation was calculated for each of the task durations. Each condition and participant had four tasks and therefore four sections where the standard deviation of the lane placement was measured. To give a just and comparable data set, four sections from the baseline scenario were used as well. The duration of these sections was twenty seconds which is the mean time to completion of a standard task. The average of the four data measures was calculated to get an average of the standard deviation for each participant. All the participants' data was summarised and a total average value of all the participant, sorted by

condition was made. This data was then compared and a statistical hypothesis test (T-test) was done to see if there was any significance between the three different driving conditions. The T-test was done in Microsoft Excel.

When analysing the lane placement for conditions D, E and F, a similar approach to condition A (baseline) was used. Since the tasks were not only done in the instrument cluster, the duration could not be seen in the log file and the shoulder camera that filmed the road ahead and the head up display did not cover some of the tasks. This made it impossible to determine when a task ended. Again the mean time for a generic task was used to get an estimation of the lane placement variation. The twenty seconds were measured from the time of task initiation, which was observed in the log file. The four sections were then in a similar way used to calculate an average of the standard deviation and lastly a total standard deviation of the lane placement for all the participants was calculated. The results can be viewed in chapter 4.5.3.

There is however a built in error when analysing data only when a specific task is performed. If the task completion time is too short the lane position value will appear unnaturally straight. To investigate if this had any effect on the lane position comparison, a corrected measuring was done. The corrected measuring took place by extracting a time segment from each condition where it was certain that a task was being performed. The mean time for completing a task in condition C was about ten seconds so therefore a ten second period was extracted from each condition. This corrected value is described in chapter 4.5.3 and discussed later in chapter 5.1.

Distance to Truck in Front

In a similar way to the lane placement calculations, the distance to the truck in front was calculated. Measuring the distance to the truck in front was done during the time that the participants were completing the given tasks. In the same way, four reference sections were taken from the baseline scenario (A).

The same was done for conditions D,E and F. Four sections, each twenty seconds after every task initiation were extracted and their average was calculated. Finally a total standard deviation was extracted.

Gaze on Road

Every timestamp creates a row of cells in the log file. With a frequency of twenty samples per second (one every fifty milliseconds), the total session time is the cell count divided by twenty.

The eyetracker returns a number on how good the eye reading is for both the right and left eye for every timestamp. The range goes from 4 (none at all) to 0 (both which eye and its gaze position). To determine how much of the participants' gaze that was placed upon the road a counter was added in excel. The percentage was calculated according to the following formula:

$$Gaze\ on\ road = 1 - \frac{number\ of\ 4's}{Total\ number\ of\ cells}$$

Equation 2 - Percentage of Gaze on Road

This was done both for the left and right eye separately as these values may differ from each other. The total number of cells refers to the active driving session. The gaze on road values was calculated for all the different conditions and the result can be viewed in chapter 4.5.5.

3.6.9 Test Summary

With the experience from the data collected and the comments from the participants, improvement suggestions were worked out. These suggestions recommend what can be done in the future and can be found in chapter 5.4.

4 Results

This chapter contains all the results that have been gathered throughout the thesis work. Any result that is too big to appear here can be found in the appendix section.

4.1 Results of data collection and analysis

This is the result of the pre study and problem identification.

4.1.1 User studies and observations

The areas that were identified as important and frequently occurring were that the drivers:

- Look away from the road when talking on the phone or navigating the radio
- Miss social interaction
- Do tasks while driving that should have been done before departure i.e. setting the GPS destination
- Lack information while driving i.e. traffic accidents, ETA
- Getting blinded by the sun, mirrors etc
- Having a bad routine i.e. missing to complete tasks before departure

4.1.2 Concept specifications

The must have criteria that the concept had to fulfil in order to get selected for further development.

No.	Need	Weight (1-5)
1	Easy to calibrate	3
2	Improve driving experience	5
3	Needs driven	4
4	Enabling communication (system)	3
5	Providing desired info	4
6	Keeping driver alert	5
7	Easy to understand	4
8	Easy to operate	5
9	Works well with other functions	3
10	Context based functions	3
11	Feels safe	4
12	Encourages use	3
13	Instills pride	2
14	Fun to use	2
15	Safe to use	5
16	Works in all conditions	4
17	Monitors driver well-being	3

Figure 18 - Concept specifications

4.2 Concept generation

To more easily understand the different kinds of gaze concepts that have been generated, three categories have been created. The categories do not take into account if a concept uses passive or active gaze interaction but instead focus on how they improve the driver situation. They are the following:

Reduce interference

Interference can be any type of factor that prevents the driver from doing his or her job. This can be blinding sunlight, drowsiness or a driver being drunk. A concept that reduces interference works actively to help the driver focus on his or her main task; driving.

Reduce human error

Human error is when the technology is working properly but the driver makes a simple mistake. A concept in this category works actively to prevent any incident that are caused by human error.

Inform safely

Inform safely is about presenting relevant information to the driver when the driver needs it and in a way that it cannot be missed. Furthermore it is important that the presented information does not steal too much attention from the road. The driver must be able to read the information without removing his or hers attention from the road.

4.2.1 Workshop 1

The many different ideas from the first workshop were summarised and written down on a sheet of paper. The main ideas can be summarised as follows:

Reduce Interference

Passive Observer

The system reduces interference and identifies when the driver is tired and acts accordingly i.e. suggests to stop for a coffee, starts to call a friend or urges the driver to make a stop for rest. The system does not drive itself but merely helps the driver to make appropriate decisions.

Reduce Human Error

Checklist

This concept is a passive system that works as a checklist for the driver. The system alerts the driver if he or she has forgotten something before departure i.e. forgotten to look at all the important gauges or to make sure that the mirrors are properly adjusted. The system prevents the driver from departing if not all the required settings have been checked.

Driver Coaching

The system monitors the driver and gives feedback to their driving, both in terms of encouragement when the driver acts safely but also as a reminder if the driver forgets to look i.e. when turning at a red light or when the driver forgets to look in the rear mirrors. The aim of this system is to improve the driver and also to encourage drivers to drive more efficiently.

4.2.2 Workshop 2

The second workshop was divided into two parts and resulted in numerous ideas and suggestions. These suggestions are presented below.

Passive

With the passive part of gaze interaction the possibility of monitoring the driver was investigated and ideas were generated with the 6-3-5 method modified to more of a 5-3-3 method. Three major areas were identified throughout the generated concepts and these are the following:

Reduce Interference

Driver Wellbeing

This concerns how a monitoring system can identify if the driver is tired or drunk. If the driver is unable to drive the vehicle the system will alert the driver of this and maybe even stop the truck if needed. Furthermore the system could be able to identify when the driver gets sun in his or hers eyes and adjust the visors to protect the driver's eyes and or intensify the display if the driver has trouble reading it and in this way managing the driver wellbeing.

Driver Recognition

The system could also be used to identify the driver's face and thus reducing the risk of having a truck stolen or having drivers tampering with driving time regulations. The information would be stored in the driver card that logs the driven time, rests etc. Each time a driver enters a new vehicle and inserts their driver card a validation of the iris is necessary to start the vehicle.

Reduce Human Error

Driver Attention

The second category concerns what the driver is doing and if the driver is doing something that puts the driver or fellow motorists in danger. This could be to alert the driver if he or she is looking away from the road for too long or if the driver is not looking on the right things. This could also include if the driver fails to look before entering an intersection or if something is forgotten before departing.

Interrupt Music

Another concept was to pause the music the driver was listening to if attention was

not given to the road and in this way reminding the driver to stay alert.

Inform Safely

Context Based Information

One of the concepts is to have context-based information that is presented to the driver when it is needed. This could be to have the lights in the cab go up when the driver is looking at the instrument panel at night and to fade when the driver is looking back at the road, this is done to reduce the amount of reflections in the windows.

Another idea was to present important information where the driver was looking to ensure that the information is not missed and yet another idea concerned presenting more information when the driver is looking at a specific point on the instrument cluster i.e. display estimated time of arrival (ETA) when the driver looks at the time.

Active

The second part of the workshop consisted of the random words method with the three categories; Triggers, Functions and Feelings. The random words were drawn from each category in three iterations and resulted in the following:

Round	Trigger	Function	Feeling
1	Scrolling	Change truck fuel economy paramters	Curious
2	Frowning	Change lights	Annoyed
3	Touchscreen double tap	Activate camera	Hungry

Figure 19 - Random Words

In every iteration the participants generated concepts that needed to incorporate all of the three categories.

Reduce Interference

Pupil Contraction Monitoring

One concept to solve this was to look at the sun visor and press a button on the steering wheel to lower it or to make the truck flash its lights once when looking at the opposing car and frowning. The concept would based on the pupil dilation and light sensors in the truck adapt the driving environment to make the driver more comfortable.

Inform Safely

Information Scrolling

Concepts for this could be looking at an icon in the instrument cluster and then scrolling a button on the steering wheel in order to extract more information about

the specific things the driver is looking at. This context based navigation would act as a natural way of presenting information only when needed and be more intuitive than today's arrow based navigation that takes attention to what context or even if the driver is looking.

Augmented Reality

Another solution was to double tap a food icon on the instrument cluster to start an augmented reality that projected the nearest restaurants in a head up display in front of the driver. This would save the hassle of finding points of interest in a map or a GPS menu stealing attention from the forward road scene.

Messages in Right Place

This concept was about presenting the message for incoming calls in an area that the driver is looking at. In this way the incoming call could not be missed and the driver would not have to look away when answering the phone.

4.3 Concept selection

With all the concepts sorted and categorised, a screening process with a lot of discussions took place. Selection was done by internal discussions but also with the input from some of the test drivers at Scania. Lastly the concepts were matched against the concept specifications and only five were relevant enough to move forwards with. The following concepts were chosen:

- Checklist to make sure nothing is forgotten
- Interactive night panel that dims when not looked at
- Head up display interaction through button press and gaze
- Eye monitoring that adapts the lighting, sun visor etc based on the eyes
- Gaze support that aids the driver to look on the road

4.3.1 Level of Gaze Implementation

The final five concepts have been placed in the level of gaze implementation graph as can be seen in Figure 20. Where the concepts are placed in the graph was determined through extensive discussions with the supervisor and the driver interaction group at Scania.

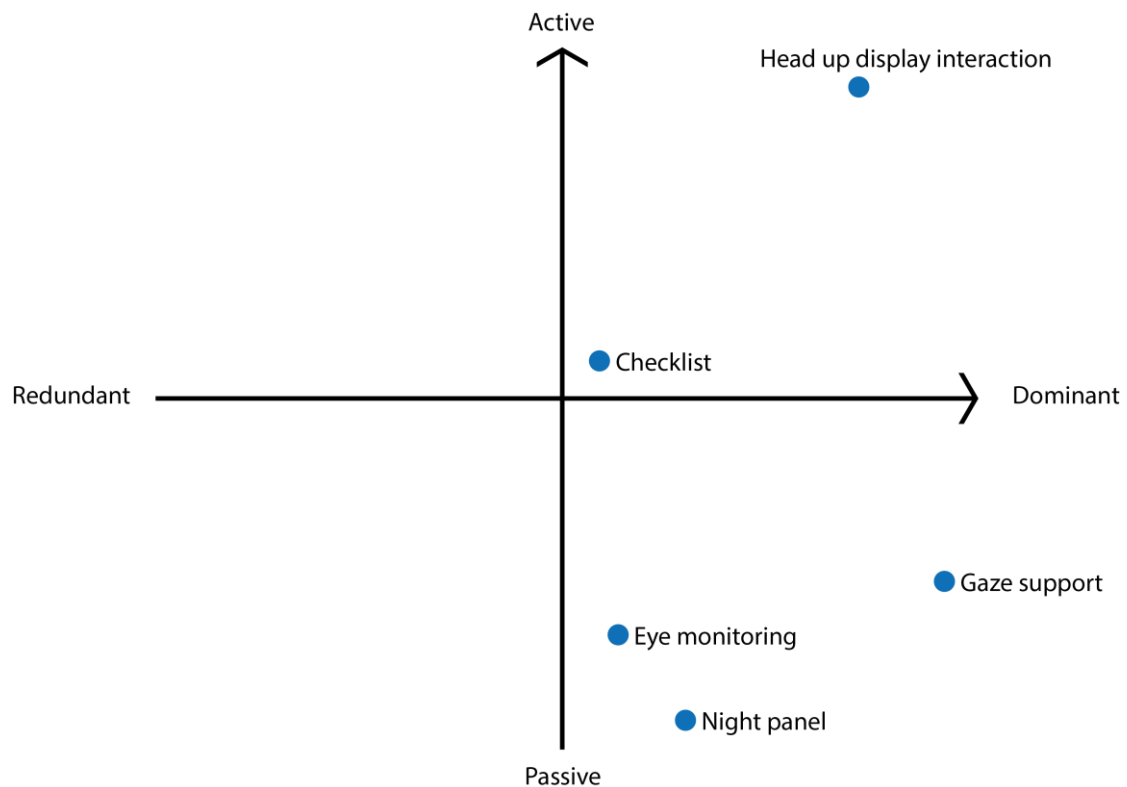


Figure 20 – Concept level of gaze implementation

All of the concepts are more on the dominant side of the graph. The two concepts that are the most dominant are the Gaze support system and the head up display interaction. As mentioned before all of the concepts are on the dominant side but the checklist, eye monitoring and night panel have a higher level of redundancy in them and do not add the same unique features as the others.

4.3.2 Concept scoring

A concept scoring took place in the way described by Ulrich and Eppinger (Ulrich & Eppinger, 2008) and it landed in the following result:

Gaze support	239
HUDI	218
Night panel	212
Eye monitoring	210
Checklist	184

Figure 21 - Concept selection result

As can be seen, the Gaze support and the head up interaction concepts scored the highest even though the other concepts came in just below the top two. The full scoring with the weighted values can be viewed in appendix 8.6.

4.3.1 Technological Feasibility

All the concepts are more or less feasible. However, both the night panel and the checklist require a larger field of tracing than is offered in the eyetracking solution that was available. One solution would have been to use multiple eyetrackers, something that the budget did not allow. These two concepts were therefore less likely to be tested in the simulator.

4.3.2 Implementability in simulator

In order to fully evaluate the proposed concepts they needed to be tested in the simulator. After discussing what was possible and not in the simulator it was discovered that the eye monitoring system would be hard to test in the simulator. There is no sun visor in the simulator and even if one was to be installed it is not controllable through a computer which makes it hard to implement in the simulator. Since the pupil diameter is a parameter that is measurable in the eyetracker, it would be possible to just test the stability of that signal. The proposed method was to blind the participants with a light and see if this could be statistically validated in the log file afterwards. Unfortunately time did not allow for such tests to take place.

The night panel concept was also hard to implement due to the fact that there is a limited amount of lights in the instrument panel and the few there are, are not controllable until some electrical work has been done. Time was of an issue and the concept could not be realised.

4.3.3 Final selection

As mentioned before it was important that the tested concept would shed light upon the possibilities in the gaze technology and to satisfy the aims and goals of the thesis. A concept that is redundant in the implementation chart is less likely to prove the unique use of gaze interaction. This assumption together with adding all the decision parameters together resulted in two final concepts.

- Head up display interaction
- Gaze support system

This is a combined result of the above mentioned reasons together with discussions at the driver interaction department at Scania.

4.4 Final Concepts

The concepts that were selected and developed further are described in detail below.

4.4.1 Head Up Display Interaction

The head up display interaction system consists of a quick menu that contains five icons for rapid access to functions or information that is used frequently. The idea is that the menu can contain any information and that the drivers are able to customise the menu as they please. The aim with this concept is to test the way of

interaction and to determine if this way of interaction is a better way of navigating information than the current solution. For demonstration and testing purposes, five functions were chosen. They are the following:

- Fuel consumption
- Brake pressure
- Speedometer
- Navigation
- Music



Figure 22 - HUDI Inactive Icons

The system is navigated by pressing and holding down a button on the steering wheel. This activates the gaze system and the menu lights up and a bar is lit below the specific icon that the driver is looking at to give feedback. When the driver wants to select an icon he or she releases the button and the highlighted choice gets selected.



Figure 23 - HUDI Activated Icons

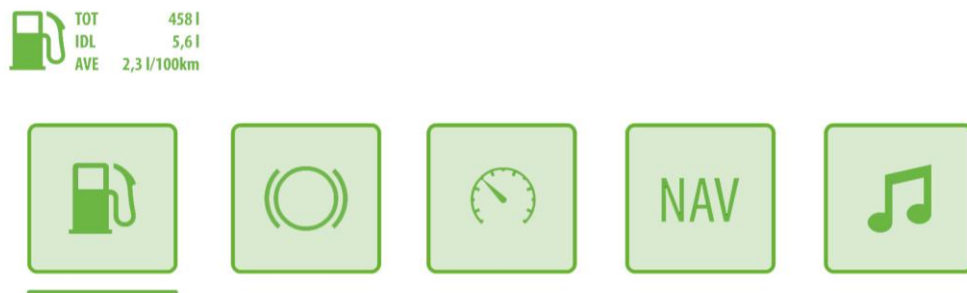


Figure 24 - HUDI Fuel Activated

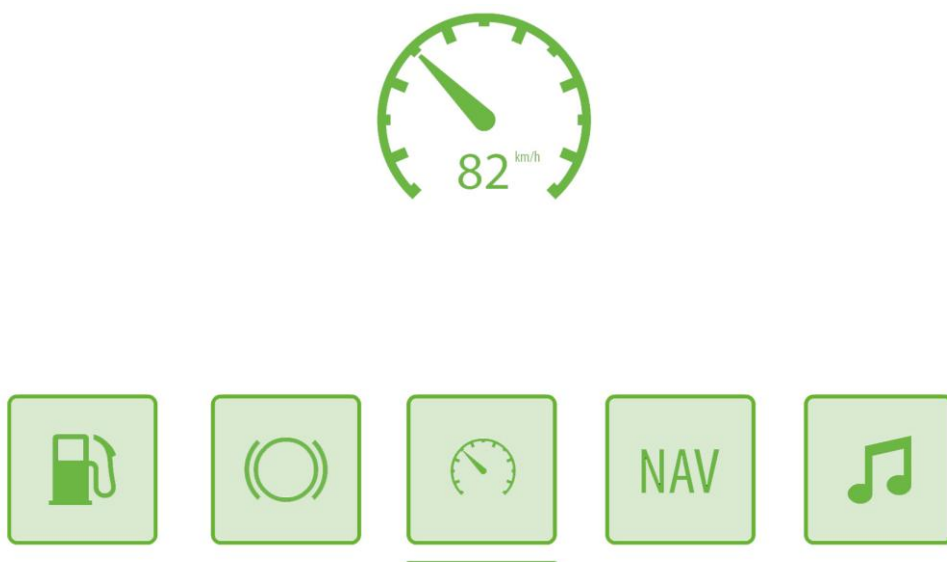


Figure 25 - HUDI Speedometer Activated

The chosen information is displayed above the quick menu. When the button is released the menu also dims down to not steal attention from the road or the presented information. When the driver wants to get rid of the displayed information, the button on the steering wheel is pressed again and the driver looks either at the icon that corresponds to that information or the displayed information and then releases the button. When this is done the information disappears. In this way the driver can quickly display and hide any information in the menu without shifting the gaze from the road.

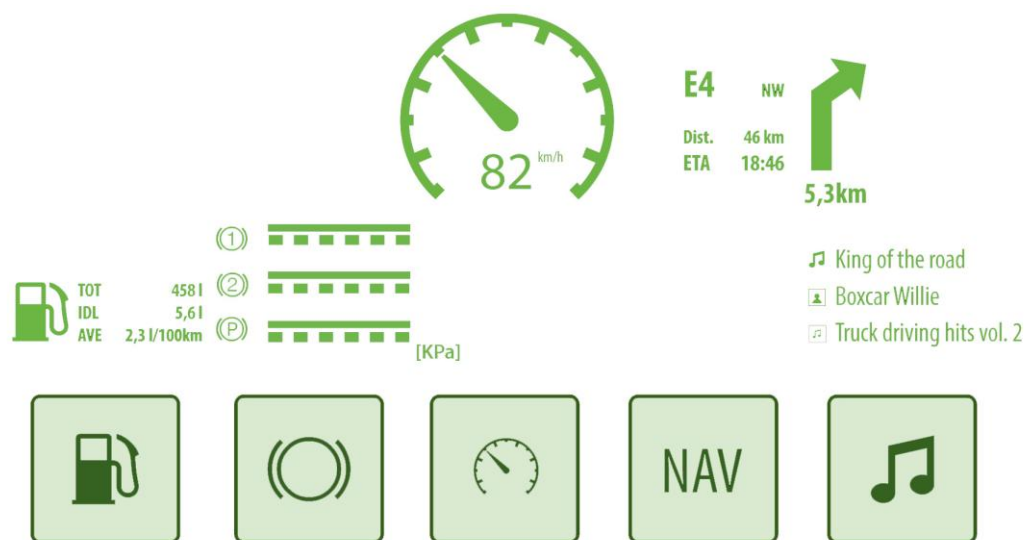


Figure 26 - HUDI All Information Displayed

The menu is mostly transparent with wireframe buttons that do not cover too much space in the forward road scene. The gaze sensitive areas in the concept are larger than the icons to ease selection even if the registered gaze is outside of the icon. This is not noticed by the user but gives a more smooth user experience.

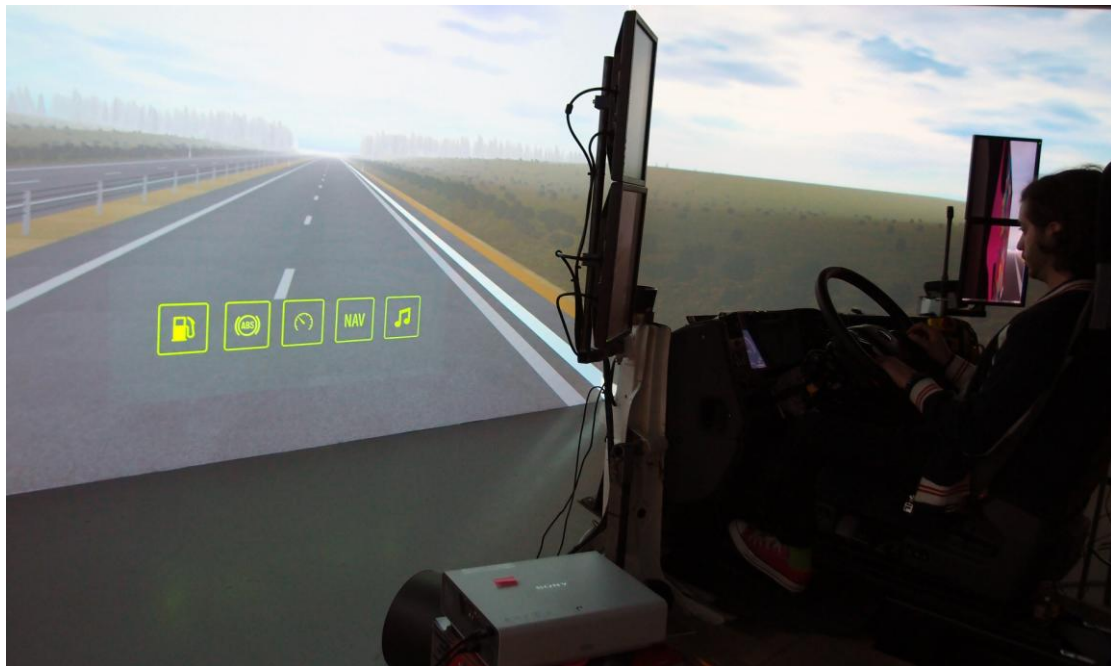


Figure 27 - HUDI system in simulator

4.4.2 Gaze Support System

Based on the National Highway Traffic Safety Administration (NHTSA) guidelines the gaze support system was developed. The system works in a way that if the gaze is placed outside of a predefined virtual box in the forward road scene or if contact with the eyetracker is lost, a warning will appear after 2,0 seconds. The warning is an orange flashing triangle inspired by the information on how the human eye works and that the eye is really quick on picking up movement in the peripheral field of view (Johnson, 2010) (Tobii Technology, 2010). If the driver does not look at the forward road scene for another 2,0 seconds the warning level will go up and a red triangle will flash in front of the driver. The red triangle is accompanied by a warning sound that alerts the driver if the driver's eye happen to be closed or if the driver does not see the first warning. There is another type of warning that appears if a driver looks at a static point for too long. If the driver looks at a specific point for more than 2,0 seconds and blue triangle will silently start to flash. Again, if the driver does not shift the gaze for another 2,0 seconds the red warning triangle together the with warning sound will appear. Both these warnings disappear as soon as gaze is registered within the predefined box again. The aim of this concept is to help drivers keep a safe viewing pattern with more time spent looking at the road than on a cell phone.

4.5 User Testing

In this chapter the results from the simulator testing is described. Participant 1 only finished half the simulator test and is therefore excluded from the test result.

Participant two finished the whole test but the data was too limited due to the fact of a bad reading from the eyetracker. Participant three did not even get a reading for the calibration process and is therefore also excluded from the test results. The reasons and effects of this will be discussed in chapter five.

4.5.1 Expectation Measures

Test ID	Scenario 1 - HUD						Scenario 2 - Gaze support					
	Condition A		Condition B		Condition C		Condition D		Condition E		Condition F	
	Before	After	Before	After	Before	After	Before	After	Before	After	Before	After
1												
2	2	1	4	4	4	2	4	3	7	5	3	6
3							2	2	4	3		
4	4	1	4	3	4	4	5	2	4	6	2	4
5	4	4	5	5	5	4	3	6	7	6	6	6
6	4	5	4	4	2	2	5	7	7	7	7	6
7	2	2	2	2	4	1	3	3	3	3	2	2
8	1	1	3	2	3	1	2	2	3	4	4	5
9	1	1	1	1	1	1	1	1	1	2	1	1
10	5	2	5	3	7	5	6	2	6	2	6	3
11	1	1	5	2	5	4	4	4	7	7	7	7
12	3	6	2	7	2	1	3	8	7	9	7	8
13	4	5	5	5	3	4	5	7	7	8	6	6
14	1	1	3	4	3	3	5	1	6	3	5	3
Sum:	30	29	39	38	39	30	42	43	58	57	53	51
Mean:	2,50	2,42	3,25	3,17	3,25	2,50	3,23	3,31	4,46	4,38	4,42	4,25

Figure 28 - Expectation Measures Scoring

It can be seen in the individual answers that a person that rated a high difficulty was more likely to rate high throughout the entire test. This may be some explanation as to why the result may look spread out. The mean is calculated based on the number of respondents in each specific condition.

The expectation measures are illustrated below. Firstly demonstrated by condition and lastly a summary of the mean for each driving condition. Please note that this scale ranges from one to ten where one is “very easy” and ten is “very difficult”. The axes are ordered opposite from Tullis & Albert and the reason is to create a scale that is more comprehensive and with a finer granularity.

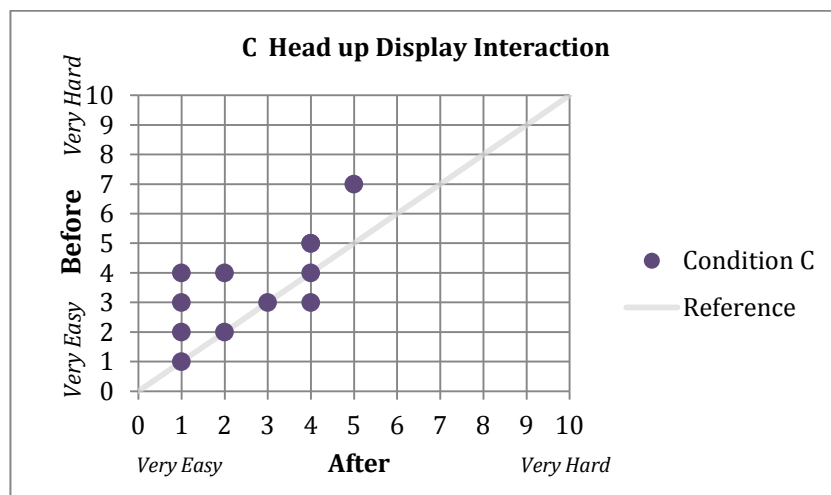
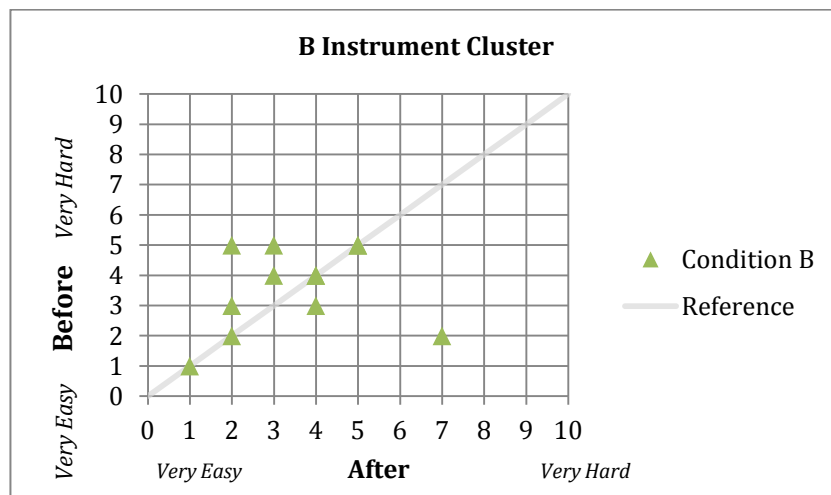
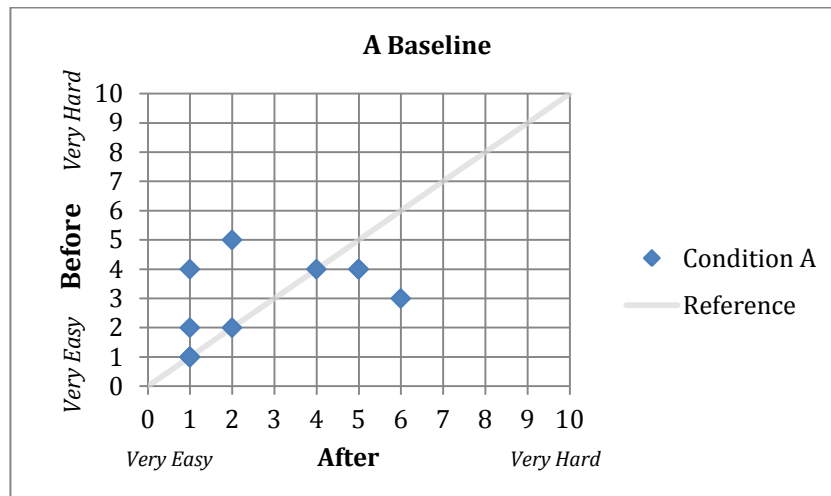


Figure 29 - Expectation Measures Condition A B C

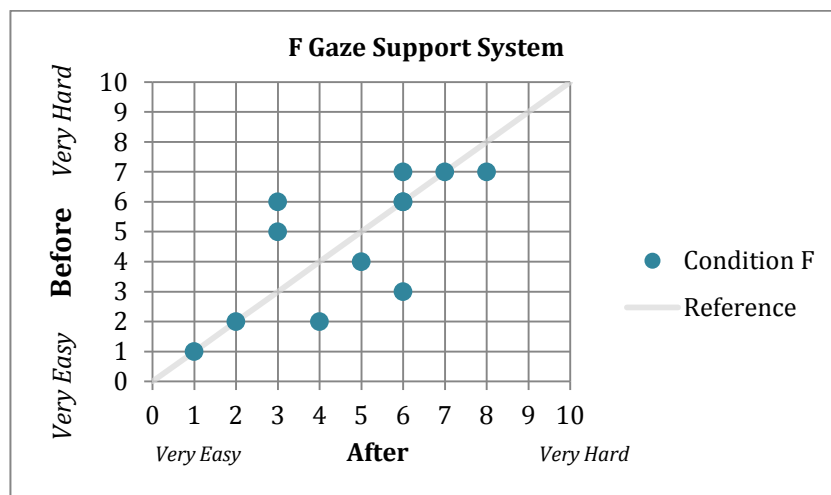
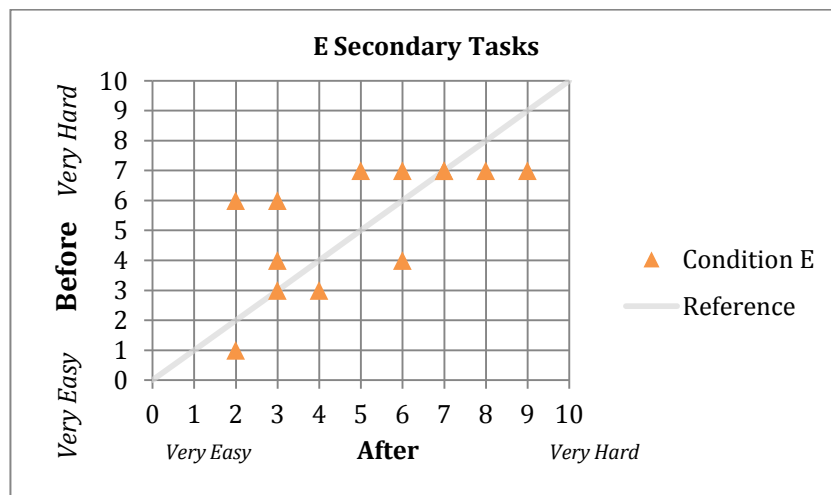
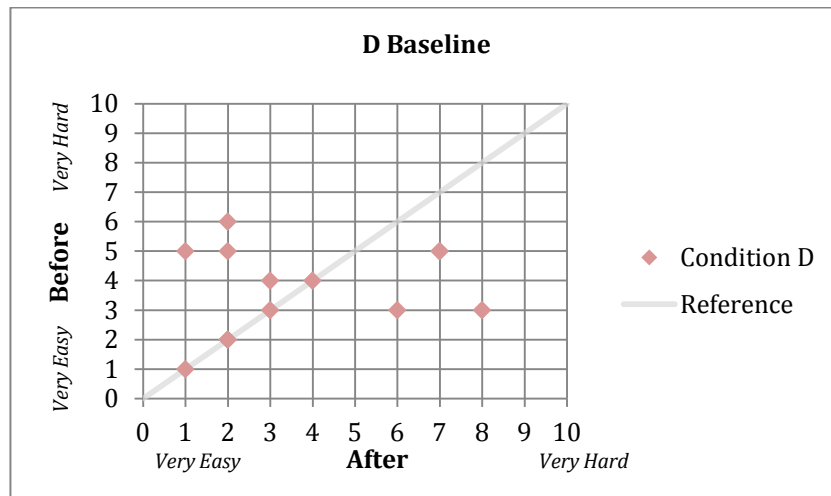


Figure 30 - Expectation measures condition D E F

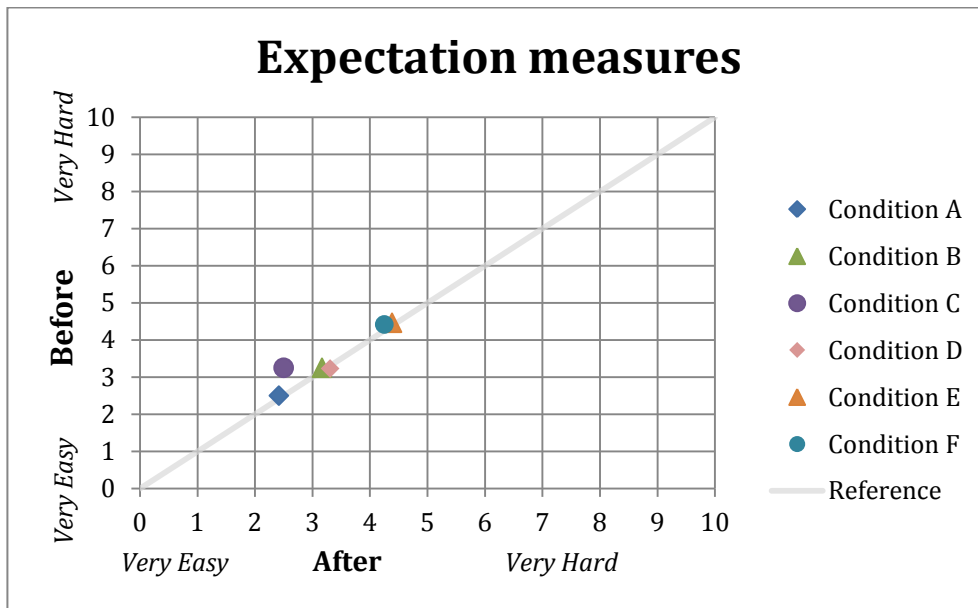


Figure 31 - Expectation Measures Mean Values

The mean expectation measures are placed very close to the centre line regarding before and after. As mentioned before it seems like a person that rated high before a test also rated high after it. In the same manner people that rated low also continued to rate low throughout the entire test. This may explain the centricity of the mean compared to the individual results. As can be seen in condition A-D they rated low both before and after the driving conditions. This places them in the “don’t touch it” area described by Tullis & Albert (Tullis & Albert, 2013). The reference line is a way to visualise if a condition in general was easier than it seemed or more difficult. A placement on the line or to the left of it indicates that the condition was as expected or more easy than anticipated. Conditions E and F placed a little higher on the scale indicating that they were more difficult than the other driving conditions putting them in between the “don’t touch it” and “great opportunity” in the expectations measures areas. Since all conditions are placed on the centre line it can be assumed that their average expected difficulty corresponds well to their experienced difficulty.

4.5.2 System Usability Scale

The results from the system usability scale questionnaires are described below. When calculating the mean, care has been taken to the participants that did not fill out the forms. The score ranges from 1 to 100 where 100 is the maximum usability score.

Scenario 1 Head Up Display Interaction

Participant	HUD										SUS score
	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	
1											N/A
2	5	1	4	1	3	1	5	1	4	2	87,5
3											N/A
4	5	1	4	1	4	2	5	1	5	1	92,5
5	5	1	4	1	4	1	4	3	4	1	85
6	5	1	5	1	5	1	5	1	5	1	100
7	2	4	4	1	4	2	4	3	2	1	62,5
8	4	1	4	1	5	1	4	2	5	1	90
9	5	1	5	1	1	1	5	1	5	1	90
10	4	2	4	1	5	3	4	2	4	1	80
11	2	3	2	3	2	2	2	2	2	4	40
12	5	1	5	1	5	1	5	1	5	1	100
13	4	1	5	1	3	1	4	1	5	2	87,5
14	5	1	4	1	5	1	5	2	4	2	90
Mean											83,41

Figure 32 - SUS Head Up Display System

The head up display interaction system was in general very well liked by the participants, something that can be seen in the fairly high score of 83,41.

Scenario 2 – Gaze Support System

Participant	Gaze Support System										SUS score
	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	
1											N/A
2	2	4	3	3	4	3	2	4	5	1	52,5
3											N/A
4	3	1	5	2	5	2	5	4	5	1	82,5
5	3	3	4	2	4	1	4	3	3	1	70
6	4	1	4	1	5	1	5	2	5	1	92,5
7	4	2	4	1	4	2	3	4	3	1	70
8	2	1	4	1	2	3	1	1	4	1	65
9	5	1	5	1	1	1	5	1	5	1	90
10	3	3	4	2	4	3	2	4	4	3	55
11	3	4	3	3	4	2	3	2	3	3	55
12	3	2	3	1	4	1	2	4	1	1	60
13	3	2	4	1	4	2	3	2	5	1	77,5
14	2	1	5	1	3	1	2	3	4	1	72,5
Mean											71,82

Figure 33 - SUS Gaze Support System

The gaze support system scored somewhat lower in the usability scale and general comments from the participants were that it was too stressful and pointed out when the drivers were making errors in a too aggressive way. Despite the comments the system scored 71,82 and the participants were in general positive to a support system given that it acted in a less irritating manner.

4.5.3 Lane Position

These are the standard deviations from the lane position measurements. To achieve a significance level of 0,05 for these three conditions a Bonferroni correction has to be done. In this case that means

$$p = \frac{0,05}{3} = 0,0167$$

Equation 3 - Bonferroni Correction Value

The tested value needs to be lower than 0,0167 in order for the comparison to be significant. Note that each of the conditions' values are collected from using the task completion time except for the baseline where an average of 20 seconds has been used in four different sections of the baseline scenario. This has been done to give all the conditions a similar amount of collected samples to make them comparable.

LP	Condition		
Participant	A. Baseline	B. ICL	C. HUD
4	0,093	0,107	0,113
5	0,162	0,086	0,069
6	0,148	0,106	0,096
7	0,107	0,098	0,040
8	0,091	0,192	0,074
9	0,073	0,195	0,014
10	0,125	0,133	0,087
11	0,114	0,112	0,094
12	0,164	0,170	0,027
13	0,059	0,080	0,043
14	0,135	0,186	0,102
Mean:	0,115	0,133	0,069

Significance:

A vs B	0,32548
B vs C	0,00539
A vs C	0,0046

Figure 34 - Lane Position A B C

As can be seen in Figure 34 there is no significance between conditions A and B. There is however a significant improvement between conditions B and C which

indicates that driving was more stable in the latter. Confusingly driving in condition C was also significantly more stable than baseline which will be discussed in chapter five.

LP		Condition		
Participant	D. Baseline	E. Secondary Task	F. Gaze support	
4	0,128	0,161	0,108	
5	0,141	0,140	0,115	
6	0,125	0,126	0,149	
7	0,141	0,120	0,087	
8	0,109	0,215	0,212	
9	0,070	0,096	0,137	
10	0,091	0,080	0,120	
11	0,124	0,239	0,138	
12	0,132	0,161	0,120	
13	0,034	0,069	0,078	
14	0,080	0,164	0,118	
Mean:	0,107	0,143	0,126	

Significance:

D vs E	0,02874
E vs F	0,22720
D vs F	0,1919

Figure 35 - Lane Position D E F

Since it was impossible to distinguish the task completion time in conditions E and F an average task completions time was used to extract the lane position data. The same 20 seconds as in the baseline scenario A was used and the timer started from when a task was initiated. This means that data was only extracted around the times when participants were performing secondary tasks. Conditions D, E and F are therefore based on the times which makes them comparable.

The lane position for conditions D, E and F are less clear and no significance can be found between any of the samples. The results however supports that the driving was the most stable in the baseline scenario (D) to get less stable when secondary tasks are added. The driving gets a little more stable in condition F when the gaze supports system reminds the participants to look at the road. The lack of significance means that no real conclusions can be drawn from this.

4.5.4 Distance to truck in front

Similar to the lane position measurements, the standard deviation was used to analyse the variation in distance to the truck in front. A Bonferroni correction was also done in this case to compensate for the three different conditions and the value is again 0,0167.

D Participant	Condition		
	A. Baseline	B. ICL	C. HUD
4	6,283	9,862	3,527
5	6,678	6,979	5,398
6	9,275	7,072	5,086
7	3,581	3,782	3,873
8	2,543	2,600	1,534
9	1,536	4,198	0,190
10	2,155	2,788	1,152
11	9,388	7,435	3,677
12	3,853	9,966	0,847
13	1,499	5,334	0,624
14	7,541	3,915	1,727
Mean:	4,939	5,812	2,512

Significance:

A vs B	0,3468
B vs C	0,0021
A vs C	0,0029

Figure 36 - Distance to Truck in Front A B C

No significance can be found between conditions A and B. There is a clear improvement in the stability to the relative position to the truck in front when looking at conditions B vs C and A vs C. The driving was the most stable when using the head up display system. The fact that the head up system was more stable than the baseline will be discussed in chapter five.

D	Condition		
Participant	D. Baseline	E. Secondary Task	F. Gaze support
4	6,279	2,555	6,428
5	5,461	5,018	3,072
6	10,732	7,653	8,613
7	2,157	12,306	10,342
8	4,558	3,808	3,858
9	2,859	6,364	4,990
10	3,145	1,904	2,987
11	4,601	4,037	5,707
12	4,721	9,668	13,538
13	3,324	2,059	2,030
14	4,992	5,878	8,443
Mean:	4,803	5,568	6,364

Significance:

D vs E	0,5419
E vs F	0,2366
D vs F	0,2079

Figure 37 - Distance to Truck in Front D E F

No significance can be found in the last three conditions for the gaze support system. A small increase in the standard deviation from baseline to condition E to even a little higher in condition F.

4.5.5 Gaze on road

Below is the registered gaze data for each scenario. The data is based on what percentage there is a good enough reading to be able to distinguish where the user's gaze is directed. Since the traceable field of the eyetracker corresponds almost exactly to the forward road scene it can be assumed that every time a gaze sample is registered, the gaze is directed towards the forward road scene.

It can be noted that participant 7 is excluded from conditions A, B and C and that participant 13 is excluded from all of the conditions. This is due to that the registered gaze data was below 30%, probably because of temporary malfunctioning of the eyetracker. The results were therefore not representative for the study as it could not be distinguished if the participant was looking at the road or not. The implications of this will be elaborated in chapter five.

GOR	Condition		
Participant	A. Baseline	B. ICL	C. HUD
4	60,89%	63,92%	64,55%
5	84,94%	84,37%	48,69%
6	76,68%	65,16%	85,58%
7	N/A	N/A	N/A
8	63,93%	63,21%	69,40%
9	88,46%	79,89%	86,98%
10	88,85%	78,81%	86,07%
11	79,21%	72,79%	78,60%
12	90,59%	80,94%	95,95%
13	N/A	N/A	N/A
14	75,09%	72,91%	77,63%
Mean:	78,74%	73,56%	77,05%

Significance:

A vs B	0,017
B vs C	0,526
A vs C	0,7173

Figure 38 - Gaze on Road A B C

There is no significance between any of the conditions when looking at the time that the eyetracker registered gaze data. There are however indications that the time spent looking at the road decreases during secondary tasks. The gaze upon road value increases with the head up display to almost the same value as in the baseline scenario.

GOR	Condition		
Participant	D. Baseline	E. Secondary Task	F. Gaze support
4	59,53%	61,08%	69,22%
5	94,57%	92,16%	81,63%
6	75,81%	69,87%	79,41%
7	82,04%	81,52%	80,14%
8	73,42%	59,48%	66,59%
9	88,13%	84,10%	79,78%
10	96,65%	67,82%	69,86%
11	78,23%	48,87%	73,99%
12	84,81%	77,75%	84,81%
13	N/A	N/A	N/A
14	50,79%	70,62%	72,97%
Mean:	78,40%	71,33%	75,84%

Significance:

D vs E	0,157
E vs F	0,171
D vs F	0,5539

Figure 39 - Gaze on Road D E F

The baseline scenario in condition D has an almost identical value as the previous baseline scenario (A) and again does the percentage drop when performing secondary tasks. When completing the secondary tasks with the aid of the gaze support system the percentage goes up again but not fully to the baseline value. The explanation lies partly in the secondary tasks being things that require the driver to look away from the road which was not the case with condition C.

4.5.6 Task Completion Times

These are the task completion times where it was possible to measure them. As can be seen the HUDI system is much quicker than navigating the instrument cluster and also with a smaller spread.

Task completion time	Condition	
	B. ICL	C. HUD
Mean:	28,14	9,34
Standard deviation:	17,63	12,49
Significance:	0,000001563	

Figure 40 - Task Completion Times

4.5.7 Standardised Results

To investigate if the increased stability in the HUDI system was due to the shorter measuring time because of the decreased task completion times, a standardised measure was made. In this measuring, four ten second intervals were extracted from each of the samples around the times when the secondary tasks were performed. The result can be seen below.

Standardised Lane Position

LPS		Condition		
Participant		A. Baseline	B. ICL	C. HUD
4		0,088	0,073	0,101
5		0,135	0,087	0,087
6		0,089	0,068	0,082
7		0,059	0,059	0,062
8		0,087	0,133	0,110
9		0,032	0,077	0,068
10		0,077	0,123	0,059
11		0,062	0,054	0,095
12		0,124	0,107	0,092
13		0,038	0,073	0,038
14		0,073	0,101	0,061
Mean:		0,078	0,087	0,078

Significance:

A vs B	0,4188
B vs C	0,3574
A vs C	0,9425

Figure 41 - Standardised Lane Position A B C

It can be seen that there is no significance between any of the measured samples. Condition B (ICL) has a slightly higher deviation than the other conditions.

LPS	Condition		
	D. Baseline	E. Secondary Task	F. Gaze support
Participant			
4	0,093	0,075	0,086
5	0,092	0,080	0,053
6	0,092	0,051	0,090
7	0,137	0,105	0,036
8	0,069	0,104	0,105
9	0,048	0,048	0,065
10	0,051	0,076	0,042
11	0,093	0,249	0,116
12	0,097	0,129	0,102
13	0,026	0,041	0,040
14	0,040	0,057	0,084
Mean:	0,076	0,092	0,074

Significance:

D vs E	0,3400
E vs F	0,2511
D vs F	0,8640

Figure 42 - Standardised Lane Position D E F

No significance can be found between the measured conditions. Conditions D and F are slightly lower than condition E that contained the secondary tasks.

Standardised Distance to Truck in Front

DS	Condition		
	A. Baseline	B. ICL	C. HUD
Participant			
4	3,256	1,741	2,848
5	2,421	3,423	2,712
6	4,810	2,737	3,842
7	1,917	14,841	1,156
8	1,442	0,728	2,048
9	0,798	1,280	1,806
10	1,224	0,544	1,248
11	5,928	3,342	4,515
12	1,726	1,443	1,926
13	0,902	1,741	0,936
14	3,821	3,013	1,907
Mean:	2,568	3,167	2,268

Significance:

A vs B	0,6497
B vs C	0,5065
A vs C	0,2876

Figure 43 - Standardised Distance to Truck in Front A B C

Yet again no significance even if the head up display concepts scores a little bit better than the other two.

DS	Condition		
Participant	D. Baseline	E. Secondary Task	F. Gaze support
4	3,555	1,449	3,087
5	2,772	1,744	2,045
6	5,288	3,230	3,744
7	0,888	1,989	12,037
8	2,397	2,626	2,343
9	1,433	1,149	1,400
10	1,972	2,218	2,223
11	2,705	3,652	2,073
12	3,055	5,256	2,731
13	2,216	1,158	1,372
14	2,418	2,465	4,312
Mean:	2,609	2,449	3,397

Significance:

D vs E	0,6977
E vs F	0,3580
D vs F	0,4776

Figure 44 - Standardised Distance to Truck in Front D E F

Condition F (Gaze Support) has the highest standard deviation even if there is no significance between the results. It can be noted that the distance to the truck in front is the most stable in condition E.

5 Discussion

5.1 Results

5.1.1 Head up display interaction system

The head up display interaction system got a very good response from the participants. No matter if the person was technical or not, the time to learn the system is measured in seconds instead of minutes. The system is also very easy to reach and to operate, something that the European commission for communities speaks of in their recommendations (The Commission of the European Communities, 2007). The high SUS score also suggests that the system is intuitive and user friendly. There are however some setbacks to its current condition. The system requires a very good calibration in order to work properly. Those users that did not have a good reading in the initial calibration were more likely to have drop out problems with the system during use. If the system dropped out the participant would have to shift their position or quickly look down on the eyetracker to get the reading back. This is not something that is acceptable in a driving situation. This happened only a few times and for most of the participants it worked flawlessly.

Looking at the lane position and distance to the truck in front there is clear significance that the HUDI system is more stable than the other concepts, even more stable than baseline. This seems a bit strange and the reason is most likely because the task completion times are much shorter, about one third, than the other secondary tasks. This creates a shorter measuring period of lane position etc which makes it look more stable. When we did the control testing using ten second intervals in all of the conditions the difference is much smaller. We cannot say that using the HUDI system is more stable regarding lane position than navigating a normal instrument cluster. What we can look at is the task completion times and the percentage of gaze on road. Even if the test group is too small to give significance it can be seen that the gaze upon road value is almost the same as in baseline and that the task completion time is about one third of navigating the instrument cluster.

With the previous statement from NHTSA that gazing away for more than 2,0 seconds increases the likelihood of crashes (National Highway Traffic Safety Administration, NHTSA, 2012) and that the peripheral field of view is really quick on noticing movements as described by Johnson (Johnson, 2010), the likelihood of preventing a road accident should be higher if the driver has their gaze on the road. The HUDI system allows for more gaze on road since it has moved up to the windscreen and it has at the same time a quicker way of navigating than other systems. This should allow for the driver to focus more on the road and therefore be safer.

One could argue that it is not fair to compare the HUDI system with navigating the instrument cluster. The instrument cluster contains almost every function in the truck and the HUDI only a few. The point of the HUDI system is not to replace the instrument cluster but to complement it. Functions that the driver use often should not be hidden in submenus but instead be easily accessible. The ability for the drivers to choose their own information for quick display instead of searching through long menus and at the same time have better view of the road is something we believe is only for good.

5.1.2 Gaze support system

Many drivers commented that the gaze support system has great potential but there are a few issues that need to be solved in order for this system to be accepted by its users. To start a warning after 2,0 seconds is something that a majority of the participants found to be stressful. Also the fact that the participants felt like someone was pointing out every wrong they made was stressful. The drivers felt like worse drivers than they actually were. The flashing triangle that was intended to catch the users eye also irritated whenever a false positive was given because of the eyetracker. A different type of warning could have been developed with the guidelines proposed by Johnson. Johnson also talks about habituation as an issue (Johnson, 2010) and this may be a problem in the future if the current time limits are kept. To reach user acceptance maybe a slower and less aggressive system needs to be developed.

The time for the fixation filter is a combination of the NHTSA research and Schleicher et. al when they talk about how fixation time correlates to cognitive processing and sleepiness (Schleicher, Galley, Briest, & Galley, 2008). The filter uses the fixation time within a circle that is defined by the system. The system tends to be more sensitive to fixation if the test subject has a good calibration. When a participant with a good calibration looked at a specific point, the reading was better and the gaze samples more centred. This posed somewhat of a problem for some participants but not for others. Participants that had a good viewing pattern and a good calibration would still get a fixation warning because of how the filter was constructed. The filter should be adapted with longer trigger times and a smaller fixation area in order for the system to work effectively. The current settings created an unnatural driving pattern and this could easily be adjusted.

There is a larger difference in the distance to the truck in front when using the gaze support system compared to when doing secondary tasks. One theory we came up with is that a person looking down in the instrument cluster menu does not see the truck in front and therefore keeps a more even pressure on the accelerator compared to a person trying to compensate by accelerating and braking more often. This is something that could be investigated in future testing.

In the future the system should adapt to what type of road that is driven. A straight highway should have different trigger times compared to a busy city or a narrow winding road in the alps. With those adjustments the system has great potential of being accepted by the drivers and therefore used.

5.1.3 Simulator test validity

There are some issues when looking at the simulator test validity. Firstly the driving experience will never be the same as in driving a real truck. Secondly the number of participants is a bit too low to find real significance in the results. The initial number was 14 participants but for different reasons the final number was only 11. Some of the participants simply did not get enough good readings to be used and this is because of the tracker. If we had used more people, even with a few percent not resulting in good data, we could have had enough to find significant results. The high repeatability of the simulator strengthens our result and makes it easier to find it relevant.

When looking at the HUDI system and in some cases the instrument cluster the process of finding task completion times was made harder since some of the participants played around with the system. This may also have affected the other tested values and we should have been clearer about not performing unnecessary tasks while driving. This also applies to the gaze support system that a few drivers wanted to test to see when it was working or not.

The secondary tasks were arranged in a way that no participant got the same order of secondary tasks. This was good as the effect of learning the interface or having complex tasks at a specific time was eliminated. It was decided though that the driving conditions were set in a specific order for every participant. The baseline scenario in condition A was more of a warm up drive in order for the participants to get to know the simulator. We kept the second baseline through condition D as a reference and found little difference between the two conditions. There is a small possibility that the study may suffer from learning how to drive the simulator. We strongly believe that this is not the case when it comes to the secondary tasks and our concepts. The reason for this being that the each of the tasks are unique and that there is no repetition. Navigating the instrument cluster is learnt in a few seconds by using the arrow keys on the steering wheel and navigating the HUDI system is even quicker to learn by holding down a button to activate the gaze interaction. The participants therefore each had a new unique task each time a secondary task was performed. The only thing that affected the test result is that some of the drivers are very familiar with the instrument cluster and therefore navigated quicker than others. The results however show that the HUDI system was a lot quicker nevertheless.

Another way to test the user attention could have been to have a different range of secondary tasks i.e. solving logical problems or counting numbers. This would have occupied the participants for longer periods of time but our goal was not to test user driving but instead, as mentioned before, to test the gaze technology.

Lane position and distance to the truck in front

When testing the lane position the significance was generally low. As mentioned before the measurement lacks in credibility since different times were used when measuring lane position. When we compensated that by using 10s periods the result changed and we found no significance. The lane position measurement is maybe not the best way of testing the quality of driving since it depends on the task completion time. A reaction time test would probably have been a better measure to the user attention and how much the drivers looked at the road. The problem there however is that there is a big issue with learning. The users would quickly learn that when doing a task, a reaction time test would appear shortly and this is something we wanted to avoid.

In a similar manner the distance to the truck in front is a measure that is not as exact as we wanted it to be. There is a possibility that the lane position and the distance to the truck in front would have been more interesting if the road was more challenging with more curves in it. The current road is very straight which means that a driver can look away for a longer period of time without swaying too much. A more curved road would have shown inattention to the road more quickly. That way the lane position and distance to truck in front measurements would have been a lot more relevant.

Gaze on road

The low number of participants and the fact that we had to remove some participants due to not having enough gaze data affected the results negatively. Even though no significance was found the gaze data behaves in a way that was expected. The gaze on road readings was the highest in the baseline conditions, the lowest when doing secondary tasks and slightly higher when using our concepts. As stated before this indicates that drivers spend more gaze on the road using the tested concepts. It would be interesting to use more participants in order to find significance in the results. There were some major issues with the tracker not registering some participants and future studies could find out if this is a frequently occurring problem with this type of eyetracker or if just was because of a bad copy. The quite wide sample spread in the registered gaze data could be that people have very different eyes and since every person is unique it is hard to build a system that can cope with all of them. That combined with our very big sample area makes the system less reliable than if a single computer screen would have been the measuring object.

5.2 Method and implementation

5.2.1 Choice of method

The choice of method was greatly influenced by different parties. Our own goal was to find a method that allowed us to make good decisions based on relevant data and at the same time made room for new and innovative ideas. Scania also had their say and wanted this project to be a test bench for a service design method. Since the

service design method is loosely defined it allowed us to pick and choose tools that we found suitable for the task at hand. This has probably affected the project by allowing for a more qualitative approach with emphasis on understanding how the drivers feel while driving and attitudes towards the developed concepts. If a more traditional product development method would have been chosen this would probably have resulted in a smaller diversity of concepts but with concept selections made from more accurate data. A parallel project has to be run in order to evaluate which is the most effective way of developing products but in our experience the service design approach suited us well.

5.2.2 Process

One could say that our process was to work iteratively with smaller iterations within the bigger iteration. In order to test a concept in the prototype phase that Legeby (Legeby, Service Design - Powerpoint presentation, 2014) described it was necessary to iterate quite a few times more with test driver feedback, finding solutions in the simulator and expert reviews again before moving on to the next phase. This meant that we worked iteratively on many levels.

Service design

When looking at the service design toolbox it should be considered just that; a toolbox. As Stickdorn et. al. states, service design should be adapted to suit the specific need of that development process (Stickdorn & Schneider, 2013). We could not agree more and think that the service design process is a good base to be complemented with methods that suit the needs. The service design process prevents the developer from being locked in a specific phase in the process and problems can be solved in a more user focused way, given the right planning. As IDEO describes it is extremely important to always stay user focused (IDEO, 2011), something we tried to follow as much as possible. During the course study we tried to follow the steps as much as we could, trying to make sure that focus was always directed to the set up goals of the project.

5.2.3 Planning

The original Gantt scheme, that was created in the beginning of the development process, was held to a great extent and did not have to be revised. The near zone planning used in the Scania way of work also helped when tasks needed to be specified more clearly in a five week period.. This visual planning is good for work overview and workload communication (Scania AB, 2013). This way of work is very close to the method described by (Tonnquist, 2010). By having a general plan for the entire project combined with a more detailed plan for the near future the delivery times have been far easier to follow. The fact that our planned phases overlapped to some extent made it easier to start a task before the other was done. This approach is in our opinion less clear but its strengths outweighs the weaknesses.

5.2.4 Context

It was challenging to find the right limitations for the project. In the same way that time has to be spent looking for relevant articles and do benchmarking it is also important to keep the time plan and to move on in order to deliver the results on time. We believe we have done a reasonably good job in identifying the main uses and implementations of gaze interaction but we are well aware that a lot more could have been done, given more time. It helped to always try to analyse if the piece of information was relevant or not. If we succeeded or not is up to the reader.

User studies

As Yin states, interviews are one of the best ways of identifying user problems (Yin, 2003) and it would have been a lot harder to do it in another way. The semi-structured type of interview described by Stanton et. al (Stanton, Salmon, Walker, Baber, & Jenkins, 2005) was used and we were very precise about following the recommendations given by Lindahl (Lindahl, 2005) with a limited amount of information beforehand and then to explain more when the interview was finished. This was done to affect the drivers as little as possible.

We are well aware that semi-structured interviews cannot be used as statistical evidence or representative for the entire target group. As Saunders et. al. mentions, qualitative interviews also makes it a lot harder to repeat the experiments (Saunders, Lewis, & Thornhill, 2009). Our goal was not to find all user problems but rather to get a general idea of what it is like to be a truck driver. This is something that we, even with our limited number of 14 interviews believe we came close to. Interviews can, if done in the right amount come just as close to identifying problems as focus groups do (Griffin & Hauser, *The Voice of The Customer*, 1993) and it was even better to conduct the interviews in the driver working environment. This way the interviewees could be observed at the same time which helped in identifying the problems. The number of drivers is limited and this also affects the identified problems.

5.2.5 Problem analysis

The user problems that have been identified probably reflect the needs in northern Europe. Had the user study been conducted somewhere else, different problems would have been identified. We believe the most important aspect of the problem analysis was to identify general traffic hazards and user attitudes to new technology. Our concepts maybe do not solve every individual problem but our aim was to find a more general solution that works on a higher level. A concept that offers quicker displaying of information can help with issues such as looking too much at the instrument cluster or not knowing where to go and therefore looking at the GPS instead of the road.

5.2.6 Development

Concept generation methods

It is easy to find concept generation methods. It is however extremely hard finding concept generation methods that create the right type of result. On one hand the method needs to make room for creative depth, on the other hand it needs to generate solutions relevant to the very limited gaze interaction technology. This is an issue that Gobble discussed and we strongly agree (Gobble, The Persistence of Brainstorming, 2014). A big issue with the first workshop was that most of the solutions did not relate to the gaze interaction technology enough. An example could be that “the concept identifies that the driver is tired and suggests to stop for coffee”. This solution has more to do with how to stop for coffee than with gaze. This created many redundant ideas since our focus was slightly off. The result from the first workshop was good but lacked somewhat in creativity. To counter this, a new set of methods were used for the second workshop. The brainwriting worked very well and the fact that the sketches can be interpreted in many different ways opened up for new and innovative concepts (Linsey & Becker, Effectiveness of Brainwriting Techniques: Comparing Nominal Groups to Real Teams, 2011). The brainstorming worked well but both the advantages of people triggering each other to create new ideas described by Wilson (Wilson, 2013) as well as the disadvantages of people loafing, described by Furnham (Furnham, The Brainstorming Myth, 2000), occurred. As Scanell describes it is more complex than it seems to create a good brainstorming session (Scanell & Mulvihill, 2012).

The random words method worked well and was easy for everyone to adopt. It was challenging to create new ideas though when all of the ideas needed to incorporate gaze interaction as a technology. It was also very important to define the task clearly enough for everyone to understand before the session started. The most challenging part of creating the workshops was to not be too specific whilst still trying to generate results in a quite narrow field. The workshops were also a bit limited in the time that was available. Since the two different workshops generated very different results we believe that the combined result is relevant and we are pleased.

5.2.7 Concept selection

There were many factors that affected the concept selection phase. Even though many concept were good solutions on their own and solved specific problems, the aim of the study was to evaluate the technology as a whole. This evaluation was an important factor that affected the choices we made. The basic sorting and screening took place through discussion and this worked well as every concept was rated with its pros, cons and how it would show the strengths and weaknesses of gaze. Using the help of the test drivers at Scania was really good as they quickly could say if something would work or not. Their input helped to sort out if something was relevant or not combined with the previous problem definition.

The concept specifications also played an important part in selecting the final concepts. The concepts were scored and rated as is recommended by Ulrich and Eppinger (Ulrich & Eppinger, 2008). Our experience is that it is very important to really think about what the scoring result means and how the winning concept would perform in a real usage situation and to think of its relevance. This is something that sometimes is forgotten when fixating too much on the numbers. It is crucial to set the right specifications in order for the concept scoring to work properly. The fact that we had so many other factors to consider probably helped with this.

The technical limitations such as us only having one eyetracker and what equipment that is available in the simulator also affected the choice. What good is a perfect concept if it cannot be tested? The simulator system development expert Matteo helped us immensely and even pushed the limits of the simulator beyond what we believed was possible. He gave good input on what concept that would provide a more relevant result than the others. Lastly the level of redundancy in the gaze concepts played an important part in the final selection. Placing the concepts in the gaze chart described in 4.3.1 really helped too in the selection. When all this information finally was weighed together with the project aims the choices were a lot easier to make.

If time would have allowed it would have been nice to test a few more concepts and even a few that tested specific solutions. This probably would have given an even better picture of the strengths and weaknesses of the gaze interaction technology.

5.2.8 Further Development

When developing the concepts further, we tried to take as much care as we could to the human scale as possible and adapt the concepts to be within visibility limits and so on (Diffrient, Tilley, & Harman, 1993). This was also one of the main reasons for moving our concepts up into a head up display. We also tried to incorporate as much of the design guideline proposed by Johnson (Johnson, 2010) and Norman (Norman, 1983) in order to have a system that speaks the user's language. Even though the system not is perfect, we believe that most users are able to learn our systems pretty quick. Initially the HUDI system icons were enlarged when the user looked at them as has been done by Miniotas et. al. (Miniotas, Špakov, & MacKenzie, 2004). This system was abandoned after discussions that it would disturb more with too much movement.

There are always minor details that need to be adjusted before a concept is fully realisable and the work from concept to simulator testing contained a lot of iterations. Last we need to add that no considerations were taken when selecting colour and icon shapes when the interface in the concepts were developed. We just picked colours and icons to show our concepts and did not research if it was the best solution, but it worked out fine anyway. That is however a factor that needs to be researched further when future work should be done. It was good that we had time for this and we got a lot of help with the pilot testing from people in the department.

The gaze support system did not get as much pilot testing and this probably shows in the SUS questionnaires. The statistical help we got from one of the statisticians in the department was of great help in knowing what to measure and how to measure it. The fact that there was a working simulator when we started saved a lot of time in setting up all of the right equipment.

5.2.9 Simulator testing

As mentioned before, the simulator is a good way of testing scenarios with a high demand on repeatability. It is also far cheaper than it would have been to install and test the gaze interaction system in a real truck. However it will never quite reflect the driving environment of a real truck. Real testing needs to be done in order to know if a system will work or not. Some of the drivers commented that the lack of feedback from the simulator made it harder to drive the way they are used to. The lack of driver feedback has probably affected the test results. The good thing is that is probably has affected all of the participants equally. The different amount of simulator experience could however have affected the results since the participants learn how the simulator works after a few driving conditions. We do not know how big this effect is but it probably is there.

A minor thing that also could have affected the simulator testing is where and how the head up information was projected. In a real driving environment the information is projected a few meters ahead of the vehicle or at bumper distance (Weihrauch, Meloeny, & Goesch, 1989) (Garrett, Bret, & Zeljko, Evaluating the Usability of a Head-Up Display for Selection from Choice Lists in Cars, 2011). In our simulator environment the information was projected on the wall at a fixed distance from the driver. This may have affected the viewing as the drivers had to refocus their eyes each time they would look at the interaction system compared to what they would have needed to do in a real truck. This may have limited the attention from the forward road scene in a way that will not be an issue in a real driving environment.

If the test was to be done again, only one baseline scenario would have been used. The two baseline scenarios initially had two slightly different setups that were changed late in the pilot testing, making them very similar. It would have been nice to be able to measure more than what was measured in this test. Maybe have an element of reaction time testing that would really prove if the driver had the attention directed forward or just the gaze, something that this test does not show. Furthermore with the knowledge we have afterwards, a better setup log file with more clear timestamps would have been created that was easier to extract data from i.e. task completion times.

5.2.10 Test analysis

The test analysis consisted of a lot of repetitive work. The active times from the conditions had to be extracted manually and the task completion times also had to be observed in the video material. This could have been avoided with some smart

planning beforehand. We also believe that the log file sample rate was too low. It is possible that information is missed with the current sample rate of 50ms per timestamp. This is a setting that is easily changed and can be up to 25ms per sample with the eyetracker as the limiting factor. We did not use the built in analysis tools from the Tobii eyetracker studio. This a powerful tool but did not quite match our preferences and more relevant data could be extracted from the log file. In a future test better use should be made of the program, maybe with areas of interest and more detailed gaze statistics. The statistics however are only relevant if the eyetracker can be trusted, something that varied probably due to the fact that we maximised the field of tracing.

5.2.11 Theoretical Framework

The process of obtaining relevant information has been both easy and hard. Areas that have a lot of research going on is generally easier to find both information supporting a statement and criticising it i.e. conducting interviews and using focus groups. When it comes to benchmarking interactive gaze systems in modern cars it has been almost impossible to find information that is relevant and not outdated since the technological advancements are taking place at a neck breaking speed. Most of the systems that are under development are of course being held a secret by the manufacturers.

5.3 Relevance

The transport industry is huge and is most likely going to expand even further in the future. With high demands on deliveries in time and a low willingness to pay, the work environment for truck drivers can be stressful. As mentioned before, many drivers often feel tired and they are unable to stop due to the strict time schedule. If our system is implemented successfully even on a small scale in this business it has the possibility to reduce the number of road accidents that are caused by drivers that are tired or not paying attention. A reduced number of road accidents is needless to say a good way to save both money and lives. Transport companies would gain from this system by ensuring that their drivers are safe and on time, a development that has already begun through GPS tracking. In the same way that many transport companies have alcohol interlock systems to ensure sober drivers this system could be used to ensure drivers with a safe driving pattern. The gaze interaction system could also work as a bridge between the systems that are operational today and the fully autonomous vehicles that will emerge in the future. Today it is unclear what the semi autonomous systems can and cannot do. A gaze system that is monitoring when the driver is aware or not could more easily alert the driver if the system is unable to deal with a traffic situation and simply not alert the driver when the system can manage on its own. This would reduce the eventual confusion with aid systems that only handles specific situations until the next generation systems appear. Since most of the road accidents that occur today are due to human error, it seems logical to try to help the drivers as much as possible until the vehicles are fully autonomous. Our system does not necessarily make the drivers better, but it gives them a better chance of seeing what is in front of them.

5.4 Reflections and recommendations

5.4.1 Gaze interaction technology

To really evaluate and get a final review of how suitable this technique is for a truck, tests should be conducted in a real environment i.e. a real truck. As Roto et. al. mentions it is important to remember that the UX of a system always is dependent on the context it is in. It is therefore crucial to test gaze Interaction in a real truck environment not only to see if the technology works but also to see how the users respond to the system (Roto, Law, Vermeeren, & Hoonhout, 2011). How the outcome of a real test would be like we can only speculate, but we think that number of external factors would affect the driving experience. This could be the rapid changes in ambient light, the driving experience in real life which is very different feeling from the driving physics in the simulator. There will probably be some issues with the calibration since objects move in a 3D environment compared to the 2D simulator environment. The independent movements from both the air suspension in the cab and the chair will cause some tough measuring problems. The device itself needs to be more accurate than it is today, we experienced that sometimes the tracker could just loose tracking. For this technique to be a part of a future truck it needs to be both feasible and trustworthy otherwise it wouldn't be safe to use when driving. The potential lies in its capacity to collect useful data from the driver and to use it in a way to help the driver to drive in a safer way. The data can easily be used to make smart functions. As the eyetracking technology keeps getting better it will not be long before the systems are good enough for real field testing. The proposed accuracy of 0,4 degrees that Tobii state they have today (Tobii Technology, 2014) is something we only experienced rarely, probably because of our slightly different setup.

Tobii Technology state that they are the best eyetracking manufacturer in the world (Tobii Technology, 2013). If this statement is true then eyetracking still has a bit to go. With the eyetracking devices out today you would have to use several devices to be able to track the whole field of sight in a truck and only then the technology can be used in its full potential.

There is also the issue with accuracy at varying illumination. Tobii's own test reports show that the accuracy and precision drop when there is less light. Since the lighting conditions in a truck cab vary from bright sunlight to almost complete darkness this is bound to have an effect on the eye tracking accuracy. The question is however if these changes in accuracy and precision are great enough for the user to notice. While setting up the simulator test the technical consultant from Tobii was asked if this would have an effect. The answer was simply that the lighting conditions in a truck are still good enough to get a reading of the user. The simulator tests were done in lighting conditions that are similar to driving at nightfall and as can be seen in the result section results were both good and bad. Given that the technology will advance regarding reading accuracy and reliability then we do not believe that changing lighting conditions will be an issue in the future.

5.4.2 Future Work

Regarding future tests we feel that trying to test new parameters like reaction time would be very interesting. When testing reaction time we speak of it as in the secondary task. It might be a good idea to have the same secondary task at all time e.g. acknowledging a short sound and which direction it comes from. Performing new tests with an updated gaze support system and tests without the system could get more accurate data and perhaps show that a support system can move the driver focus towards the road and in the end show a safer driving with significance. As mentioned before when doing further work on the concepts or work in general regarding this technique, considerations should be taken when choosing colours, icon and symbols in the interface.

The test design could also be improved. We felt that we were missing timestamps of different things that happened in the test and if we had time stamps in the collected data we could have easier managed all the data and selecting the right data when analysing.

Just like in the case of autonomous driving there are some dilemmas regarding who's responsible it is in the event of a crash. First the technology has to be 100% reliable before you would even consider implementing it to a production vehicle.

To be in the forefront of this field we feel that it is necessary to do further research. We believe that this technology will become more and more relevant and implemented in the coming vehicles of all sorts. The news of car manufactures that are testing this technology have all ready been spread in media, so the assumption of that is that all the bigger manufactures have done research in to this technology for awhile. By saying this, we would like to encourage Scania to continue with doing research in this field and we wish them good luck.

6 Conclusions

6.1 Research question 1

What information is possible to extract from the driver using eye tracking while driving on the highway?

There are a number of different types of information that are possible to extract from each eye. They are the following:

- 2D gaze position (X,Y)
- 3D gaze position (X, Y, Z)
- Eye validity
- Pupil dilation
- Timestamp

With this information it is possible to extract more data such e.g. fixations if the user is staring at a specific point and for how long. Saccades which is the time between each fixation can be analysed to see what the user missed. It is also possible to define areas of interest that can be really useful when studying user viewing behaviour. By applying more filters and smart programming, deeper information like sleepiness detection and detecting people under the influence of drugs and alcohol is possible to do.

6.2 Research question 2

How does an active gaze interaction interface affect the driving experience?

The head up display interaction system has proven to be an intuitive and easy to use concept and promotes a safer driver. Based on the system usability scale the system will be well adopted by the users if calibrated correctly.

6.3 Research question 3

Can gaze utilise the peripheral field of view in order to optimise the information flow?

Yes it is possible with the help of gaze interaction system to change the driving so more focus is directed towards the road with help of the peripheral field of view. However it is important that the system is context sensitive and reflect the current driving situation so it will not become disturbing or a stress factor. Information can be placed and presented for the driver so it cannot be missed.

6.4 Research question 4

Is gaze Interaction a technology that is suitable for the truck driver environment and interface?

There are a few issues that need to be solved for the implementation to be appropriate. It needs to be clear when the gaze system is active and when it is not. This is to make sure that the user understands when the system can be trusted. The reliability of the eyetracker needs to be high enough to guarantee tracing for every kind of user, even people with glasses. For the system to be adopted, a quick and easy calibration process is crucial. To cover the entire field of view in the truck, more than one device or an updated version is needed. If these issues are dealt with, gaze interaction is a technology with the potential of creating a safer driving environment.

6.5 Project objectives and aims

“The aim of this project is to investigate the gaze interaction technology and to assess if it is a technology that is suitable for the truck driver environment. This will be done by studying the target user group before developing one or more innovative concepts that are then tested and evaluated in the Scania driving simulator.”

Information in the field of eyetracking has been investigated and a user study involving both distribution truck drivers and long haulage drivers has been conducted. This led to a problem definition and concept specifications. With that in mind two creative workshops were held that resulted in numerous new concepts. After concept selection, two concepts remained that were refined enough to answer the set up research questions. These concepts were then tested in the driving simulator and the data has been analysed and evaluated.

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8 Appendix

This chapter contains all the information and illustrations that are too big or not directly connected to the result of the report.

8.1 Questions to the truck drivers

Frågor

Arbetet

1. *Hur länge har du arbetat som förare?*
2. *Har du kört något annat än distribution?*
3. *Finns det några vanliga faktorer som dyker upp i arbetet som förhindrar det du gör?*
4. *Finns det på samma sätt något du gör som förenklar din arbetsuppgift?
Kanske något som du kommit på själv eller en egen rutin.*
5. *Vilka situationer kräver mest fokus?*

Störfaktorer

6. *När i själva körningen upplever du att det är som stressigast?*
7. *Vad är det som bidrar mest till det, finns det något specifikt moment?*
8. *Hur hanterar du en stressig situation?*
9. *När upplever du att det är som lugnast?*
10. *Vad är det som bidrar mest till det, finns det något specifikt moment?*
11. *Kan det bli för lugnt?*
12. *Finns det stunder då du känner trötthet?*
13. *Vad gör du om du känner dig trött(beskriv)?*
14. *Kan du känna att du blir uttråkad ibland?*
15. *Vad gör du om du känner dig uttråkad(beskriv)?*

Funktioner

16. *Vad tycker du om mängden funktioner i dagsläget?*
17. *Vad är överflödigt?*
18. *Saknar du någon speciell funktion?*
19. *Finns det någon funktion som fanns förr men inte finns längre som du saknar?*
20. *Vilken/vilka funktion skulle du rangordna som viktiga för dig (utöver funktioner som rör körning, koppling, gas, styrning o.s.v.)?*
21. *Hur går det till när du använder dessa? (förklara step by step)*

22. När du ligger och kör på motorvägen, är det några speciella funktioner som används då? (berätta/förklara)
23. Skiljer sig detta mycket mot annan körning? Om ja på vilket sätt, förklara
24. Skiljer det sig på vilka funktioner som används vid nattkörning mot körning på dagen? (inte helljus och dylikt)
25. "What if" Be föraren beskriva vad som händer om man trycker på t.ex. en magisk knapp (tanken är att föraren ska spåna fram något helt fritt som löser eller utför något man verkligen vill/önskar).
26. Har du något som du skulle vilja tillägga eller ta upp? Är det något du tycker jag borde veta allmänt om körningen

Observationspunkter

- Hur ser en typisk dag ut rent praktiskt, du kommer till jobbet... sen?
- Hur utför förarna vissa bestämda uppgifter?
- Något som händer väldigt ofta?
- Vilken ordning, antal steg etc.
- Vilka funktioner medför att förarna behöver titta bort från vägen?
- Uppkommer det under några specifika situationer?
- Kolla efter tillfällen då föraren löser en uppgift/situation på ett "eget" sätt
- Hitta ställen där föraren arbetar kring ett problem utan att tänka på det d.v.s. gör en sak på ren rutin(hänger kanske lite ihop med punkten ovan).
- Vilka situationer kräver mest fokus?
- Om föraren måste ändra på sin körning drastiskt eller byta körstil p.g.a. av utomstående faktorer (när en sådan situation dyker upp ställa frågor kring detta).

Att tänka på

Fråga varför, i de fall som det förklaras varför något görs på ett visst sätt. (Tänka 5X varför)

8.2 Summary of user studies

The drivers:

Type of driving:

- Long haulage 4
- Distribution
- dangerous goods
- fluid contents
- International

Limiting factors:

- Traffic accidents
- Truck malfunction
- Cars in traffic when they enter highway 2
- Sometimes too calm during night drives
- Heavy traffic (stressful)

Stuff that helps:

- All the functionality in the truck
- Cruise control
- Distance radar

When tired

- Raise radio volume (loud) 5
- Lower window to let in fresh air 4
- Chewing gum
- Stop for powernap 3
- Screaming
- Singing along to the radio
- Talk in my cell phone 4

When bored

- Talk in my cell phone 4
- Listening to music/radio 5

Truck functions that are good/used

- Signalling lights
- Retarder 3

- Cruise control 5
- Distance radar 5
- Lowering and raising the truck
- Trailer brake (when docking) 3
- Hill stop 2
- Scania light switches
- AUX connectors
- Touch radio
- Scania's info computer
- Driver assistance game
- Drivers own GPS 3

Truck functions that are unnecessary

- Lowering of the boogie
- "Volvo is like a spaceship"

Truck functions that are wanted

- Heating of the wind shield wipers 2
- Camera for reverse parking
- Wild life detector
- Oil level indicator 2
- Electronic sun blockers
- Sun blocker on the right door (doesn't exist today)
- Rear window
- More options on cabin lights (like Volvo cars) to dim and choose colour
- Com radio 4
- Better lighting in the cab, especially reading light
- Always internet
- HUD 3
- Traffic info
- Reliable info (from the truck and from around)
- Want info on ETA, actual speed (not three different ones, färdskrivaren, IPC & gps), distance left & speed trap's
- Some type of fuel consumption info (like the one Volvo have)
- Dead angle warning
- More detailed error messages
- Truck recognition, seat settings, Bluetooth etc (like wifi recognize a mobile)

Comments

- HUD would be nice but will it work in sunny conditions
- It is like driving with autopilot
- A1 from bundestrassse to Rade is extremely boring!
- Night driving has lower visibility
- Gets angry from other traffic
- "The horn is important"
- "The Scania horn is placed in a good position"
- All the syncing and the adjustments to the interface for the driver (such as gps radio etc) can be done before departure! It is just a bad routine to do it while driving!
- "I don't want any info on the wind screen"
- "There is no function that makes me look away from the road" (But he did, several times)
- Sometimes it gets boring, need more stimuli
- "A Volvo you ride along, a Scania you drive"
- "The social part of trucking is really important"
- "think it is smart to have info presented on a small screen but not on a HUD"

Observations

- Retarder is used
- Distance keeper is used
- Drinking water
- Steering with knees when eating or drinking
- Had to brake for cars entering the highway
- Use their own GPS for extra info (speed, arrival time, time left driving)
- A lot of lane change warnings (wiggly)
- Problems with syncing the Bluetooth (mobile to handsfree)
- Did most of the settings and preparations before he took off.
- Talking on phone
- People often look at their mobile

Looking away from the road when:

- Adjusting chair
- Talking in cell phone 3
- SMS 5
- Surfing
- Drinking 2

- Eating food 2
- Snus
- Operating radio 3
- Adjusting their own GPS 2
- Syncing mobile with handsfree and interface in general
- Fleet manager app 2

Identified problems

- Looking away from the road ex. Talking in mobile
- Missing social interaction
- Do stuff while driving that should been done before taking off (adjusting the radio)
- Lack of info while driving
- Getting sun in the eye's, mirrors etc
- Bad routine (checklist like pilots before leaving)

8.3 Information from MODAS

Situation	Function	Score
Highway	Communicate	54
Highway	Lane placement	54
Highway	Obstacle avoidance speed control	54
Highway	Presetation of subsystems status	54
Highway	Road layout speed control	54
Highway	Road surface speed control	54
Highway	Route	54
Highway	Speed stability	54
Highway	Steering (fine, gross, lane stability)	54
Highway	Travel time (driven)	54
Highway	Travel time (estimated)	54
Highway	Vehicle following speed control	54
Highway	Vehicle stability	54
Highway	Visibility	54
Highway Low traffic	Communicate	36
Highway Low traffic	Lane placement	36
Highway Low traffic	Obstacle avoidance speed control	36
Highway Low traffic	Presetation of subsystems status	36
Highway Low traffic	Road layout speed control	36
Highway Low traffic	Road surface speed control	36
Highway Low traffic	Route	36
Highway Low traffic	Speed stability	36
Highway Low traffic	Steering (fine, gross, lane stability)	36
Highway Low traffic	Travel time (driven)	36
Highway Low traffic	Travel time (estimated)	36
Highway Low traffic	Vehicle following speed control	36
Highway Low traffic	Vehicle stability	36
Highway Low traffic	Visibility	36
Highway	Hill speed control	27
Highway	Information about road situations	27
Highway	Information about traffic situations	27
Highway	Route change	27
Highway High traffic	Communicate	27
Highway High traffic	Lane placement	27
Highway High traffic	Obstacle avoidance speed control	27
Highway High traffic	Presetation of subsystems status	27
Highway High traffic	Road layout speed control	27
Highway High traffic	Road surface speed control	27
Highway High traffic	Route	27
Highway High traffic	Speed stability	27
Highway High traffic	Steering (fine, gross, lane stability)	27
Highway High traffic	Travel time (driven)	27
Highway High traffic	Travel time (estimated)	27
Highway High traffic	Vehicle following speed control	27
Highway High traffic	Vehicle stability	27
Highway High traffic	Visibility	27

8.4 Workshop 1

8.4.1 Problem scenarios

"Pga. olycka på den större väg som du kör på blir du omdirigerad till en mindre väg du inte känner till så väl. I och med att det är en mindre väg som du kör på blir trafiken betydligt tätare och risken för kö och andra saker som kräver mer fokus mycket mer påtaglig. Eftersom du inte känner till vägen du kör på så väl vet du inte riktigt när du kommer att vara framme eller om du kommer att hinna till din destination i tid. Du vill också veta om och i sådana fall när man kan ansluta till den större vägen igen"

"Unga förare saknar ofta körrutin vilket kan göra dem till osäkra förare (ex inte gjort allt klart innan avfärd så som ställt in gps, anslutit mobilen i bilen eller synkat den via bluetooth osv.) medan erfarna förare ofta kan vara fast i dåliga rutiner (ex köra utan bälte, fokusera för lite på vägen osv.). Hur kan man hjälpa förare skapa och bibehålla bra rutiner?"

"Du närmar dig slutet på ditt arbetspass och känner dig uttråkad och trött. Att få ringa en polare och snacka lite skulle vara riktigt gött och det skulle göra den sista biten av resan mycket skönare. Det jobbiga är bara att du kör i Tyskland och där är de väldigt hårda mot just användandet av mobil när man kör, så du vågar inte riktigt chansa."

"Du kör distributionsbil vilket innebär många stopp och ingen lastkaj är den andra lik. Du ska backa in till en ny lastkaj som du inte varit vid tidigare och har svårt att få en tillräckligt bra helhetsbild av kajen. Du vill inte springa in och ut ur lastbilen hela tiden och känner att speglarna inte räcker till alla gånger."

8.5 Workshop 2

Tidsåtgång: ca 1-1,5h

Deltagare: ca 3-5 pers per grupp.

Material: pennor, post-its, papper.

Att tänka på: dokumentera kontinuerligt med kamera.

Genomförande

Välkomna alla!

- Dela in i grupper (eller om alla placerar sig jämnt vid borden)
- (Prata om exjobbet och vårt syfte)
- Prata om gaze och hur tekniken fungerar, vad den kan mäta och hur det funkar. Passiv & Aktiv

Passiv

Förklara tydligt att man ska försöka knyta samman det som kan mätas från ögat med funktioner, funktioner där användaren inte är aktiv utan funktionen triggas automatiskt.

Deltagarna ska idégenerera under 3 minuter per runda och tanken är att man ska ha genererat 3 idéer var. Dessa skickas vidare i gruppen.

Efter ett helt varv väljs en favorit på varje papper ut av varje "pappersägare" sedan skickas idéerna till nästa grupp. Sedan upprepas första delen en gång.

Idégenerering sker med penna och post-it lappar. När allt är klart presenteras den "färdiga" idén.

Aktiv

Här förklara man istället att det handlar om aktiva funktioner som användaren är med och påverkar.

Deltagarna hjälps åt att generera de tre olika kategorierna till metoden random words. Kategorierna är:

- Gazeinteraktion
- Funktioner
- En känsla

Varje kategori genereras under två minuter.

En lapp av varje slumpas fram och deltagarna genererar nu själva och under tystnad så många idéer de kan. Brainstormingen itereras så många gånger vi hinner.

Sammanfatta workshopen och låt deltagarna kommentera eventuella tankar och förslag.

8.6 Concept scoring

			Checklist		Night panel		HUDI		Eye monitoring		Gaze support	
No.	Need	Weight (1-5)	Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score
1	Easy to calibrate	3	3	9	3	9	3	9	3	9	3	9
2	Improve driving experience	5	4	20	5	25	4	20	4	20	3	15
3	Needs driven	4	3	12	4	16	4	16	4	16	5	20
4	Enabling communication (system)	3	2	6	2	6	4	12	2	6	2	6
5	Providing desired info	4	3	12	3	12	5	20	2	8	3	12
6	Keeping driver alert	5	2	10	1	5	3	15	3	15	5	25
7	Easy to understand	4	3	12	5	20	3	12	4	16	4	16
8	Easy to operate	5	4	20	5	25	3	15	5	25	5	25
9	Works well with other functions	3	3	9	4	12	5	15	3	9	5	15
10	Context based functions	3	2	6	4	12	4	12	4	12	5	15
11	Feels safe	4	4	16	3	12	3	12	4	16	3	12
12	Encourages use	3	3	9	3	9	4	12	3	9	3	9
13	Instills pride	2	2	4	4	8	5	10	2	4	3	6
14	Fun to use	2	2	4	3	6	4	8	2	4	1	2
15	safe to use	5	4	20	4	20	3	15	4	20	5	25
16	Works in all conditions	4	3	12	3	12	3	12	3	12	3	12
17	Monitors driver well being	3	1	3	1	3	1	3	3	9	5	15
			184		212		218		210		239	

8.7 Expectation measures questionnaire

Upplevd svårighet Gaze support

Ringa in det alternativ som du anser passar bäst.

Körfall 1

Hur svårt uppskattar du att körfall 1 kommer bli?

1 2 3 4 5 6 7 8 9 10

Hur svårt upplevde du att körfall 1 var?

1 2 3 4 5 6 7 8 9 10

Körfall 2

Hur svårt uppskattar du att körfall 2 kommer bli?

1 2 3 4 5 6 7 8 9 10

Hur svårt upplevde du att körfall 2 var?

1 2 3 4 5 6 7 8 9 10

Körfall 3

Hur svårt uppskattar du att körfall 3 kommer bli?

1 2 3 4 5 6 7 8 9 10

Hur svårt upplevde du att körfall 3 var?

1 2 3 4 5 6 7 8 9 10

Kommentarer:

8.8 System usability scale questionnaire

1. Jag använder gärna varningssystemet

1 2 3 4 5
instämmer inte alls Instämmer helt

2. Jag drar mig för att använda varningssystemet, det är onödigt komplicerat

1 2 3 4 5
instämmer inte alls Instämmer helt

3. Jag tycker varningssystemet är lätt att använda

1 2 3 4 5
instämmer inte alls Instämmer helt

4. Jag behöver ta hjälp av en kollega eller någon teknisk person för att kunna använda varningssystemet

1 2 3 4 5
instämmer inte alls Instämmer helt

5. Jag tycker att varningssystemet är tydligt

1 2 3 4 5
instämmer inte alls Instämmer helt

6. Jag tycker att varningssystemet är inkonsekvent och ologiskt.

1 2 3 4 5
instämmer inte alls Instämmer helt

7. Jag kan tänka mig att de flesta skulle använda varningssystemet

1 2 3 4 5
instämmer inte alls Instämmer helt

8. Jag tror att många tycker varningssystemet är besvärligt att använda

1 2 3 4 5
instämmer inte alls Instämmer helt

9. Jag känner mig säker på hur jag ska använda varningssystemet.

1 2 3 4 5
instämmer inte alls Instämmer helt

10. Jag behövde lära mig mycket innan jag kom igång med varningssystemet

1 2 3 4 5
instämmer inte alls Instämmer helt

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